

What Constrains Insolvency in Property Insurance?

Market Discipline, Capital Regulation, and Catastrophe Exposure

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September 2, 2026

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Abstract

This paper disentangles the roles of capital regulation and credit ratings in mitigating insolvency risk in the U.S. property insurance market. I first investigate the mechanism through which capital requirements affect the insurance market. Using an instrumental variable approach that exploits a 2017 policy change as a quasi-experiment, I find that a \$1 million increase in required capital leads insurers to hold \$3.098 million more in capital and to raise insurance prices by 0.095 percentage points. These results reveal a direct trade-off between financial stability and consumer affordability. To further explore the underlying mechanisms, I develop a structural model in which insurers make capital, reinsurance, and pricing decisions in a competitive market with limited liability and exposure to catastrophic risks. Counterfactual analyses show that tightening capital requirements improves solvency but raises prices. In the absence of capital regulation, the model predicts that the insolvency rate would increase by 0.07 percentage points, while insurance prices would decline by approximately 1.2%, accompanied by a reduction in capital holdings. A third counterfactual considers a market

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I am deeply grateful to my advisors, Paul Schrimpf and Gorkem Bostanci, for their invaluable guidance and support. I also thank my committee members, Katherine Wagner and David Green, for their insightful comments. I would like to thank Torsten Jaccard and the attendees of my presentations for their insightful questions and valuable feedback. I appreciate the helpful feedback from participants and discussants at 19th CIREQ PhD Students' Conference. All errors are my own.

without capital regulation but with high credit-rating salience. Greater consumer emphasis on credit ratings does not reduce the insolvency rate. Instead, insurers respond by increasing their reinsurance holdings to improve credit ratings, which raises the assets required to remain in the market and, consequently, increases market concentration. Overall, the findings underscore that capital regulation remains crucial for sustaining market stability, as heightened rating salience alone cannot fully substitute for its stabilizing effects.

In recent years, the frequency and severity of extreme weather events have increased, leading to substantial natural and economic losses. In 2022 alone, global natural disaster damages surpassed \$270 billion, with the United States accounting for approximately \$140 billion of that total. Property insurance plays a vital role in post-disaster recovery by facilitating reconstruction efforts. However, catastrophic events are also a major driver of insurer insolvencies. For example, following Hurricane Andrew in 1992, eleven insurance companies—ten in Florida and one in Louisiana—became insolvent. Because insurers operate under limited liability, they can declare bankruptcy to avoid excessive losses. When insolvencies occur, State Guaranty Funds intervene to compensate affected policyholders. The combination of limited liability and government-backed bailouts underscores the importance of insurer solvency as a central issue in the property insurance market.

Two primary mechanisms safeguard the solvency of property insurers: market-based discipline and regulatory oversight. Market discipline is administered through credit ratings, which evaluate the financial strength of an insurer and provide vital signals to consumers and reinsurers. Regulatory oversight is enforced through capital requirements, which serve to mitigate moral hazard and risk-seeking behaviors. These regulations also ensure a liquidity cushion to absorb unexpected losses, thereby protecting policyholders. However, the core activities of U.S. insurers are not considered to pose significant systemic risk (a conclusion from Cummins and Weiss (2014)). In an environment with low systemic risk and a robust market discipline mechanism like credit ratings, the appropriate stringency of capital

regulation in the property insurance market could reduce distortions from costs of capital. This idea is bolstered by the empirical observation that most property insurers hold capital levels more than the required amount. This “capital puzzle” raises fundamental questions about whether regulatory capital requirements or credit ratings primarily constrain insurer solvency.

This paper, therefore, seeks to disentangle the roles of these two mechanisms by addressing the following questions: What are the primary constraints on insolvency in the property insurance market, and to what extent can heightened credit rating salience substitute for capital regulation in ensuring solvency while maintaining insurance affordability? Recent debates on U.S. capital regulation have focused on whether the current static, formula-based framework should be replaced by a model-based approach that more accurately reflects insurers’ risk exposures. Before such a transition can be meaningfully evaluated, however, it is essential to understand how existing mechanisms—credit ratings and regulatory capital requirements—jointly influence firms’ solvency risk. In particular, we must assess the extent to which capital regulation constrains insolvency risk given the presence of credit ratings, which themselves provide market-based discipline. The static, formula-based framework refers to the risk-based capital (RBC) ratio described in Section 1, while the model-based approach relies on predictive models of future losses to determine the level of capital consistent with a target default probability. This paper provides the necessary groundwork for that broader policy discussion by examining the interaction between credit ratings and capital regulation in the property insurance market.

In this paper, I examine how capital requirements and credit ratings interact to shape insolvency risk in the U.S. property insurance market. The paper addresses three central questions: (1) how capital regulation affects insurer solvency and market outcomes, (2) what role capital regulation plays when credit ratings serve as an important source of market discipline, and (3) whether increasing credit rating salience among consumers can effectively

substitute for capital regulation in promoting market stability.

I begin by analyzing the mechanisms through which capital regulation influences the insurance market using a reduced-form empirical approach. The analysis centers on a key regulatory trade-off: while tighter capital requirements enhance solvency, they may also increase prices and reduce insurance affordability. To identify this effect, I exploit an exogenous 2017 policy change as an instrument for required capital, addressing the endogeneity of regulatory capital levels. Following the 2017 policy change, property insurers were required to use a 1-in-100-year scenario model to assess catastrophe exposure when calculating required capital. The results show that a \$1 million increase in required capital leads insurers to raise held capital by \$3.098 million, accompanied by a 0.095 percentage point rise in insurance prices. The increase in capital on hand appears large because the regulatory capital threshold is 200% (see Table 1). Moreover, since capital on hand is stochastic and depends on market valuations, insurers have an incentive to maintain excess capital. These findings highlight the sensitivity of insurers' balance-sheet decisions and pricing strategies to regulatory constraints.

Building on this reduced-form evidence, I develop a structural model in which insurers optimally choose capital, reinsurance, and pricing under competition, limited liability, and exposure to catastrophic losses. On the demand side, I specify a logit model in which consumers' insurance choices depend on price and credit ratings. On the supply side, insurers jointly determine capital, pricing, and reinsurance decisions while facing both market discipline and capital regulations. Insurers can choose whether to remain in the market by comparing the expected returns from participating in insurance markets with their outside option: withdrawing from the insurance market and investing the available capital in financial markets. If insurers remain in the insurance market, the model allows them to earn investment returns on collected premiums, reflecting the importance of float as a source of insurers' income. Insolvency triggers state intervention through the State

Guaranty Fund, which collects post-event assessments from consumers to pay policyholder claims—introducing an endogenous insolvency cost into the model. Social welfare is measured as the aggregate of consumer surplus and producer surplus, less the costs associated with insolvency.

I first construct the moment conditions and estimate the demand system using instrumental variables. Since price levels are unobserved, I recover the baseline prices and the price elasticity parameter from observed price changes during the demand estimation. To improve the statistical power of the estimation, I group firms according to their initial size, as measured by written premiums. On the supply side, I estimate most components directly from data, including the credit rating function, required capital function, loss-rate and asset-return distributions, and reinsurance-related parameters. The credit rating and required capital functions are estimated using linear models to avoid the potentially erratic functional forms generated by nonlinear specifications, which would complicate the computation of counterfactual analyses. Linear models are chosen based on their robust out-of-sample predictive performance. The remaining unobserved element—the random shocks in insurers’ insolvency decisions—is estimated structurally within the supply system using the model’s first-order conditions and observed insolvency outcomes. Due to the nature of the model, the random shocks to insolvency are set-identified (i.e., bounded) among deeply solvent firms, while they are point-identified among marginally solvent and insolvent firms. Accordingly, in the counterfactual analyses that follow, I draw the random shocks to insolvency from the fitted empirical distribution estimated among marginally solvent firms.

During the optimization process of the counterfactual analysis, since an insurer’s price and the opponents’ prices jointly determine the insurance quantity, in each iteration, insurers simultaneously determine their prices, asset allocations, and reinsurance strategies, given competitors’ choices. The process converges to a Nash equilibrium when no insurer has an incentive to deviate from its strategy. In the first counterfactual analysis, I inves-

tigate how tightening the capital regulation threshold affects market outcomes. Consistent with the reduced-form findings, stricter capital requirements strengthen solvency but raise prices. This counterfactual analysis explores the mechanism of capital regulation by evaluating its impact on the trade-off between solvency, market stability and affordability. In a second counterfactual, I isolate the contribution of capital regulation in a market already disciplined by credit ratings. Eliminating capital requirements increases the insolvency rate by 0.07 percentage points while lowering insurance prices by about 1.2%. Firms also reduce their asset holdings significantly if there is no capital requirement. This suggests the absence of regulation induces greater risk-taking behavior. The third counterfactual analysis examines a market without capital regulation but with high credit rating salience. When consumers place greater weight on firms' credit ratings, the insolvency rate does not decline. In response to heightened credit-rating concerns, firms increase their reinsurance holdings, which is also associated with a modest increase in asset holdings and greater market concentration. Although consumer and producer surplus increase following the removal of capital requirements, the insolvency costs estimated under both definitions in this paper likely understate the true social costs of insurer failure. These estimates do not capture administrative costs, disruptions from delayed or unpaid claims, difficulties obtaining replacement coverage, losses to creditors and guaranty funds, or potential spillovers to other insurers. Moreover, because property insurers are important participants in bond markets, large-scale failures could trigger asset sales and increase financial-market volatility. Thus, the measured gains in consumer and producer surplus may be offset by these unaccounted costs, preventing us from concluding that eliminating capital requirements and relying solely on market discipline would improve social welfare. Taken together, these findings underscore the stabilizing role of capital regulation, even when credit ratings provide substantial market discipline or consumers are highly attentive to insurers' ratings.

This paper contributes to the literature on the roles of capital regulation and credit ratings in the property and casualty insurance industry. The National Association of In-

insurance Commissioners (NAIC) defines the Risk-Based Capital (RBC) ratio—the primary component of insurers’ capital regulation—as a tool that grants regulators the legal authority to intervene in financially distressed insurance companies. Risk-Based Capital (RBC) ratio is a ratio between actual capital and required capital. However, the RBC ratio alone is not intended to serve as a stand-alone measure of financial solvency. Grace et al. (1998) find that RBC ratios are less effective than the Financial Analysis Solvency Tools (FAST) in predicting insurer insolvency. The Financial Analysis Solvency Tools (FAST) is a series of analytical ratios and scoring used by U.S. state insurance regulators to assess the financial condition and solvency of property and casualty insurers. Nevertheless, the combination of RBC ratios and FAST indicators provides stronger predictive power than FAST alone. Similarly, De Haan and Kakes (2010) show that insurers’ actual capital holdings are driven more by their underlying risk characteristics than by regulatory solvency requirements, leading most insurers to hold substantially more capital than required. In parallel, other studies examine the role of credit ratings. Basten and Kartasheva (2024) evaluate how regulators and credit rating agencies monitor and discipline the insurance industry’s exposure to natural catastrophe (NatCat) risks. They find that many insurers accept rating downgrades and subsequently increase their risk-taking. Building on this literature, my paper focuses on the interaction between capital regulation and credit ratings, aiming to disentangle the distinct roles of capital regulation play in maintaining solvency within the property insurance market. To the best of my knowledge, this is the first paper that separates the roles of capital regulation and credit ratings using structural estimation.

Secondly, this paper contributes to the extensive literature on the effects of capital requirements in insurance markets. Prior research has established a clear link between capital adequacy and market outcomes. For instance, Eastman and Kim (2023) find that stringent capital requirements lead to increased insurance premiums. Similarly, Gron (1994) shows that an unexpected decrease in financial capacity results in higher prices and profitability. The literature has explored these dynamics across various regulatory landscapes. Using

variations across countries, Barbu et al. (2024) find that stricter regulations correlate with reduced ownership of traditional insurance products. In the context of the U.S. life insurance industry, Tang (2022) investigates the competitive dynamics among state jurisdictions. The study finds that while states may lower capital requirements to attract life insurers, this can lead to increased default risks and lower prices, with the associated costs borne by consumers in other states. Furthermore, research has delved into how capital regulations influence the specific investment and operational decisions of insurance companies. Becker et al. (2022) analyze a reform that removed capital requirements for mortgage-backed securities, finding that insurers were subsequently more likely to retain these assets after they were downgraded. Additionally, Niehaus (2018) find that capital flows within insurance groups are responsive to regulatory capital levels, with insurers holding lower risk-based capital receiving greater internal capital contributions. There are some papers about financial constraints of P&C insurers rather than capital regulation. Ge and Weisbach (2021) and Ge (2022) find that when P&C insurers become more constraint, they will switch to safer bond holdings and lower premiums to generate revenue. Sastry et al. (2023) find that under-capitalized and under-diversified insurers can have spillover effects on mortgage markets. While the aforementioned studies provide crucial empirical insights, this paper builds upon this literature by combining a theoretical model with structural estimation. This approach allows for a direct and quantitative examination of the fundamental trade-off between insurer solvency, financial market stability, and the affordability of insurance for consumers.

This paper contributes to the literature on optimal capital requirements in the insurance market. Goussebaïle and Louaas (2022) conducts a theoretical welfare analysis of solvency regulation in the context of catastrophe insurance. Boonen and Jiang (2023) examines the Pareto-optimal reinsurance problem under solvency constraints imposed on reinsurers. The theoretical model in Charpentier and Le Maux (2014) highlights that government-provided insurance becomes more attractive when insurer insolvency risk is taken into account during insurance purchase decisions. Cummins et al. (1993) provides a qualitative

discussion of optimal capital requirements. In contrast to these studies, this paper develops a simple theoretical framework and employs structural estimation to quantitatively assess the optimal threshold for capital regulation. By combining theory with empirical analysis, this paper provides a well-supported assessment of optimal capital requirements, grounded in both structural modeling and observed data.

Finally, this paper contributes to the literature on capital requirements under limited liability. The most closely related studies come from the banking literature. Corbae and D’Erasmus (2021) analyze the impact of regulatory policies on bank risk-taking and market structure, developing and estimating a rigorous theoretical model of bank capital requirements. Malherbe (2020) shows that the capital requirement that restores investment efficiency varies over time—being tighter during booms and looser during recessions. Davydiuk (2017) proposes an optimal dynamic capital requirement that balances efficient lending with liquidity provision, finding that the optimal policy depends on economic growth, credit supply, and asset prices. Building on insights from the banking literature, particularly Corbae and D’Erasmus (2021) and Malherbe (2020), this paper develops a model of capital requirements for insurers operating under limited liability.

In this paper, I begin by reviewing the institutional background in Section 1. I discuss data in Sections 2. I then use a policy change to examine the trade-off introduced by capital regulation between solvency and affordability in Section 3. I find that while capital regulation reduces insolvency in the insurance market, it simultaneously increases the price of insurance. In Section 4, I develop a simple supply and demand model of the insurance market to analyze firms’ choices regarding pricing, asset holdings, and reinsurance under limited liability. Section 5 presents the estimation of both the demand and supply sides of this structural model. Based on these estimates, I conduct counterfactual analyses in Section 6. The results indicate that while credit ratings are influential, the absence of capital regulation would result in higher insolvency rates and increased market concentration in some

scenarios. Even in markets where consumers are highly sensitive to insurers’ credit ratings, capital regulation remains indispensable for maintaining market stability. Finally, I conclude and offer a discussion of the implications.

1 Institutional Background

This section provides institutional background on capital regulation, credit ratings, reinsurance, and state guaranty associations, offering context that will aid readers in understanding the empirical analysis and theoretical models presented in the subsequent sections.

1.1 Capital Regulation

Regulation of the insurance industry in the United States is conducted at the state level. To promote uniformity and establish best practices, state insurance regulators collaborate through the National Association of Insurance Commissioners (NAIC).

A key tool for monitoring insurer solvency, developed by the NAIC, is the Risk-Based Capital (RBC) framework. Insurers are required to maintain capital levels commensurate with their specific risk profiles. Solvency is primarily assessed using the RBC ratio, which compares an insurer’s total adjusted capital to its required Risk-Based Capital. The Risk-Based Capital ratio is calculated with the following formula¹:

$$\text{RBC Ratio} = \frac{\text{Total Adjusted Capital}}{\text{Authorized Control Level Risk-based Capital}}$$

Total adjusted capital refers to the amount of capital a property and casualty insurer holds after all required adjustments. It is expressed as:

$$\text{Total adjusted capital} = \text{Surplus as regards policyholders} + \text{Adjustments}$$

¹For more details, please read <https://content.naic.org/insurance-topics/risk-based-capital>

Authorized Control Level Risk-based Capital (ACL RBC) is the regulatory capital requirement and is calculated using the following equation.

$$\text{Authorized Control Level Risk-based Capital} = 0.5 \times \text{Total Risk-based Capital}$$

Total Risk-Based Capital (RBC) is a standardized minimum capital requirement for insurers, determined by the inherent risks in their assets and operations. The Total Risk-Based Capital (RBC), developed by the National Association of Insurance Commissioners (NAIC), is stochastic and depends on various risk categories, including asset risk, underwriting risk, and business or credit risk, to establish the required capital level. This framework establishes several action levels based on the ratio of the company's Total Adjusted Capital (TAC) to its calculated Authorized Control Level (ACL) RBC. Each level triggers specific regulatory responses:

Table 1: Risk-Based Capital (RBC) Action Levels and Corresponding Regulatory Actions

RBC Ratio	Regulatory Action
150% to 200%	Company Action Level: An insurer falling into this range must submit a comprehensive financial plan to the insurance commissioner detailing how it will improve its RBC ratio.
100% to 150%	Regulatory Action Level: At this level, a regulator is required to perform an examination of the insurer and issue a corrective action order specifying necessary improvements.
70% to 100%	Authorized Control Level: If an insurer's ratio falls within this range, the state commissioner is authorized, but not required, to take control of the company, which can include rehabilitation or liquidation.
Below 70%	Mandatory Control Level: When the ratio is below this threshold, the commissioner is obligated to seize the insurer to protect policyholders.

Notes: This table summarizes RBC action levels and regulatory actions.

Insurers are also subject to minimum capital requirements. Minimum capital requirements establish a baseline level of capital that property insurers must maintain to support their financial solvency. While minimum capital requirements provide a uniform regulatory

floor, RBC requirements allow the required level of capital to vary with the insurer's risk profile.

1.2 Credit Rating

Credit rating agencies evaluate an insurer's creditworthiness primarily from the perspective of debt investors and counterparties. Their objective is to provide an independent, market-informed assessment of the insurer's financial strength and ability to meet future obligations. These evaluations are generally conservative, as they assess natural catastrophe risks separately and earlier than capital regulations, taking into account both current financial conditions and potential future exposures. Similar to capital regulation, credit rating agencies assess capital adequacy by considering the insurer's exposure to risks associated with underwriting net premiums, asset holdings, subsidiaries, and loss reserves. Major credit rating agencies active in the insurance market include A.M. Best, S&P Global, Moody's, and Fitch.²

In my theoretical model, credit rating will affect consumers' choice of insurance. My estimation results suggest that consumers are more willing to choose an insurance company with higher credit rating.

1.3 Reinsurance

To diversify their underwriting risks and lower the capital requirements they must maintain, insurance companies obtain reinsurance from reinsurance firms. In real world, credit rating may affect price of reinsurance. However, because of data limitation, I use a price index which is uniform across insurers to measure the price of reinsurance. This data limitation constrains the model, as it restricts the credit rating channel to influence equilibrium only

²For more details, please read *Best's Credit Rating Methodology*, *Insurer Risk-Based Capital Adequacy-Methodology and Assumptions* by S&P Global, *Property and Casualty Insurers Methodology* by Moody's, and *Prism* by Fitch.

through its effects on demand. Consequently, insurers in the model have weaker incentives to improve their credit ratings.

1.4 State Guaranty Association

In the U.S., a state-based system of insurance guaranty associations protects policyholders and claimants if an insurance company becomes insolvent. These nonprofit organizations, mandated by state law, are composed of all insurance companies licensed to do business in a particular state. When an insurer is deemed insolvent, the state guaranty association steps in to continue coverage and pay outstanding claims up to statutory limits. Funding for these payments is primarily sourced from the remaining assets of the failed insurer and through “post-insolvency assessments” levied on the member insurance companies. The amount each solvent insurer is assessed is typically based on its share of premiums written within the state. In many states, these assessed insurers are permitted to recover a portion of these costs over time through offsets on their assessments from consumers. For example, after the insolvency of Florida-based insurer St. Johns Insurance Company in 2022, the Florida Insurance Guaranty Association (FIGA) stepped in to pay out covered claims to affected homeowners.

1.5 Development of Capital Requirement in Property Insurance Market

I use a 2017 policy change to evaluate how capital regulation trades off solvency and affordability. Prior to 2017, the risk exposures R_{5A} and R_{Cat} were grouped under the same category—“underwriting net written premiums risk”. R_n represents risk exposure when measuring required capital. However, starting with the 2017 reporting year, insurers with significant exposure to hurricane and earthquake risk have been required to report R_{5A} as “underwriting net written premiums risk net of catastrophic risks” and R_{Cat} as “catastrophe

risk” separately. While the RBC ratio is generally calculated using static formulas, the R_{Cat} component is uniquely assessed using a model-based approach. Specifically, catastrophe models are used to estimate the predicted exposure to catastrophic risk. These models employ scenario-based analysis, typically assuming a 1-in-100 year event, which reduces reliance on historical smoothing and increases required capital to better reflect potential future losses. Wildfire risks continue to be included for disclosure purposes and are still not in “catastrophe risk” when calculating required capital. For the development of capital regulation before 2017, please refer to Appendix [8.3](#).

Credit rating agencies began incorporating catastrophe risk into their capital adequacy models well before regulatory authorities did. For instance, A.M. Best introduced a catastrophe adjustment to the numerator of its capital ratio prior to 2003.

2 Data

The empirical analysis in this paper draws on administrative data from the National Association of Insurance Commissioners (NAIC), the policy-setting organization for state insurance regulators in the United States. The NAIC data are at the individual company level. State-level data on natural disasters are obtained from the Spatial Hazard Events and Losses Database (SHELDUS), while information on rate changes is sourced from the System for Electronic Rates & Forms Filing (SERFF). To measure reinsurance price, I use the Guy Carpenter Global Property Rate on Line Index, which is common to all insurers and varies only over time. Except for natural disasters, insurers’ price changes, and the reinsurance price, all other variables are constructed from NAIC statutory filings. The analysis focuses on product lines that the NAIC identifies as exposed to natural disasters: fire, allied lines (including water damage), homeowners multiple perils, commercial multiple perils (non-liability portion), earthquake, and farm owners multiple perils. Government-supported or provided crop insurance (multiple peril crops, private crops) is excluded from the sample. A company’s

market share within a given state is computed as the ratio of its direct written premiums to the total direct written premiums in that state for the relevant business lines.

I want to study how capital regulation affects solvency and price. Since the required capital can be endogenous and cause issues for identification, I use instrumental variable regressions. The instrumental variable used in this study is a regulatory change implemented in 2017, which requires insurers with significant exposure to hurricanes and earthquakes to calculate catastrophic risks separately from other underwriting risks when determining their Total Risk-Based Capital. The regulation also mandates the use of scenario-based modeling to estimate future losses. Switching from historical loss data to scenario-based modeling reduces historical smoothing and increases insurers' required capital. I define the treated group as insurers that had exposure to hurricanes and earthquakes prior to 2017. Table 9 lists the regions included in the treated group. Certain states are exempt from this catastrophic risk regulation. Firms are assigned to the treated group if they underwrote policies in the lines of earthquake, homeowners, allied lines, farm owners, or commercial property (non-liability) in catastrophe-prone areas. In contrast, firms with no exposure to these business lines or whose underwriting activities were confined to states not listed in Table 9 are classified as the control group.

The primary sample spans the years 2014 to 2023 and is restricted to property insurers. I define property insurers as firms that wrote insurance in the lines of fire, allied lines, homeowners, earthquake, farm owners, or commercial property (non-liability). Table 10 presents summary statistics. Columns (1) to (3) report the mean, standard deviation, and median of the full sample, respectively. Columns (4) to (7) show the means of the treated and control groups before and after the 2017 regulation.

“Total Adjusted Capital” refers to the total surplus held by an insurer, while “Direct Written Premiums” represent the total premiums received—commonly used as a proxy for firm size in the insurance industry. The treated group consists of firms that underwrote

property insurance related to natural disasters—such as homeowners, commercial property, fire, allied lines, farm owners, and earthquake insurance—prior to 2017. The control group comprises firms without such exposure before 2017. The regulatory change took effect in 2017. The variable “**treatment**” in Table 10 is a binary indicator denoting whether a firm belongs to the treated or control group.

Firms that entered receivership before 2017 are excluded from the sample, as they do not fall into either group. The RBC ratio is defined as the ratio of Total Adjusted Capital to Authorized Control Level Risk-Based Capital, the latter being the minimum regulatory capital required. The surplus represents the capital buffer above the insolvency threshold. All financial variables, except for price change, treatment, and the RBC ratio, are reported in millions of dollars. Price change and the RBC ratio are expressed in percentages. All variables are winsorized at the 95% level.

According to the summary statistics in Table 10, the treated group experienced a substantial increase in regulatory risk-based capital and surplus following the reform, indicating stricter capital requirements and improved solvency post-regulation. Additionally, treated firms tend to be larger in size compared to those in the control group. To address this discrepancy, the empirical analysis in Section 3 includes firm fixed effects and controls for firm size.

3 Capital Regulation Trades off Solvency and Affordability

In this paper, I study the mechanism through which capital regulation affects insurers’ solvency and the insurance market. This section provides reduced-form evidence to illustrate the empirical relationship between capital regulation, solvency, and affordability. The subsequent sections develop a simple theoretical model in which insurers choose capital, pricing,

and reinsurance under limited liability. I then use this model to conduct counterfactual analyses. In the first counterfactual exercise, I tighten capital regulation to evaluate the trade-off between solvency and affordability within the theoretical framework. Empirically, this section examines the trade-off using the 2017 policy reform as an instrumental variable. As detailed in Section 1, the 2017 reform required insurers with substantial exposure to hurricane and earthquake risks to separately report catastrophe and other underwriting risks in their regulatory risk-based capital calculations. A key change introduced by the reform was the shift from historical loss estimation to scenario-based modeling, which increased capital requirements for affected insurers. The identifying variation arises from differences in insurers' exposure to catastrophic events, particularly hurricanes and earthquakes.

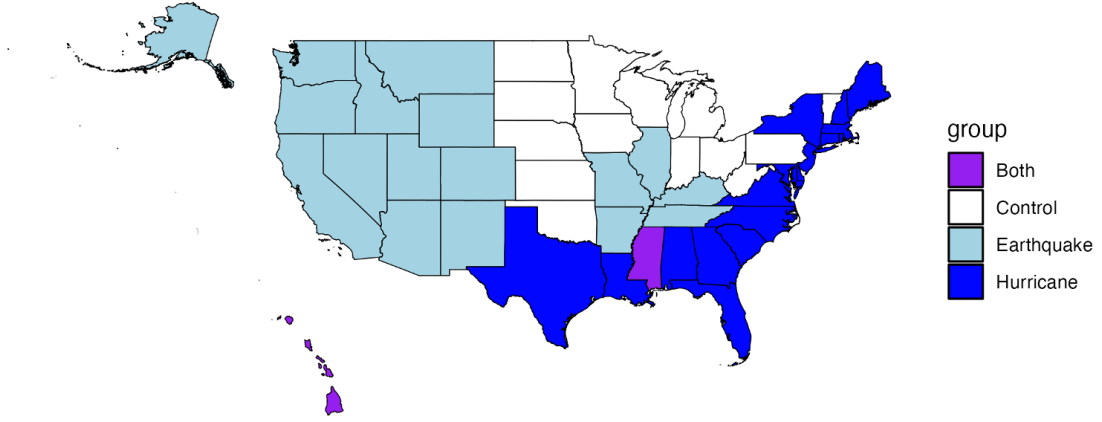
To identify the causal effects of this reform, I employ a difference-in-differences (DID) strategy combined with an instrumental variable (IV) approach. The first stage of the IV regression, which utilizes the DID framework, is specified in the following equation.

$$y_{jt} = \beta_0 + \beta_1 \text{treated}_j \times \text{reform}_t + \zeta_j + \zeta_t + \varepsilon_{jt} \quad (1)$$

j indexes insurers, and y_{jt} denotes the dependent variable, which is the regulatory risk-based capital. Treated_j is a dummy variable equal to 1 if insurer j wrote at least one policy with exposure to hurricanes or earthquakes in disaster-prone areas (as shown in Figure 1) before 2017, and 0 otherwise. The sample is restricted to property insurers that wrote property insurance during the sample period. Property insurance includes the following product lines: fire, allied lines, earthquake, farmowners, homeowners, and commercial multi-peril. ζ_j represents insurer fixed effects, and ζ_t represents time fixed effects.

Figure 1: Catastrophe-prone Areas

US States Colored by Treatment



Notes: This figure depicts catastrophe-prone areas in the United States. The light blue denotes states exposed to earthquake risks. The navy blue areas denote states exposed to hurricane risks. The purple areas denote states exposed to both earthquake and hurricane risks.

The following equation shows the IV regression.

$$y_{jt} = \beta_0 + \beta_1 RBC_{jt} + \zeta_j + \zeta_t + \varepsilon_{jt} \quad (2)$$

The endogenous variable is RBC_{jt} , representing the regulatory risk-based capital. The outcome variable, y_{jt} , includes both the price change and the surplus (i.e., actual capital held). The surplus serves as a proxy for the insurer's distance to insolvency, as greater capital holdings reduce the likelihood of insolvency. The instrumental variable is defined as the interaction term $treated_j \times reform_t$.

Table 2 displays the results for the first stage and reduced form of our analysis. The variable of interest, $treated \times reform$, serves as the instrumental variable for the endogenous risk-based capital, as specified in Table 3. The first-stage results, presented in Column (1) of Table 2, show that the policy change led to a \$3.032 million increase in regulatory risk-based capital. In the reduced form, the reform is associated with a \$9.393 million increase in surplus, which measures the distance to insolvency. Concurrently, the price change increased

by 0.289 percentage points, suggesting that property insurance became more expensive post-reform. The dynamics illustrated in Figure 2 are consistent with the scale of these regression results. To address potential weak identification, I examine the first-stage F-statistic. The obtained value of 139.403 is significantly higher than the critical values provided by Stock and Yogo (2005), ensuring that the instrument possesses sufficient explanatory power to identify the causal effect of required capital on our outcomes.

Table 2: First Stage and Reduced Form

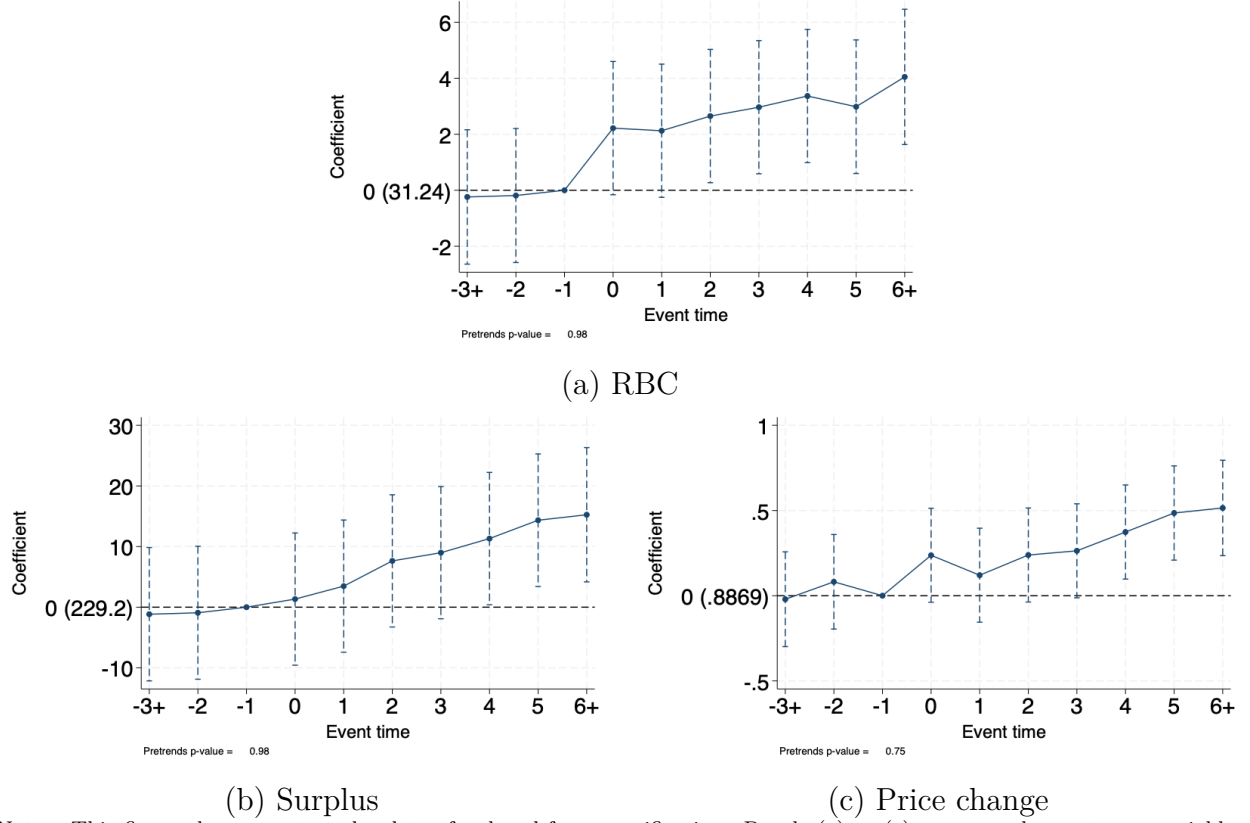
	<i>Dependent variable:</i>		
	required capital	surplus	price change
	(1)	(2)	(3)
treated $\times$ reform	3.032*** [28.452] (0.589)	9.393*** [203.721] (2.686)	0.289*** [1.147] (0.069)
Observations	16,655	16,655	16,655
R ²	0.0442	0.0493	0.0861

Note:

*p<0.1; **p<0.05; ***p<0.01

Notes: This table shows regression results of the first stage of instrumental variable regressions. The predictor treated $\times$ reform is a difference-in-difference type instrumental variable. Variable required capital will be the endogenous variable in the IV regression. Except for price change, the unit of other variables is a million dollar. The unit of price change is percent. Values in square brackets are mean values in the sample.

Figure 2: Event study plots of reduced-form specification



Notes: This figure shows event study plots of reduced-form specification. Panels (a) to (c) correspond to outcome variables “Regulatory capital”, “capital on hand”, and “price change”. Results are consistent with Table 2. Except for price change, the unit of other variables is a million dollar. The unit of price change is percent.

Table 3 reports the main results of the IV regressions. The instrumental variable is the interaction term $treated \times reform$. The outcome variables are the surplus and the price change. The results indicate that a \$1 million increase in capital requirements leads to a \$3.098 million increase in surplus, implying that insurers become less likely to face insolvency. As shown in Table 10, the RBC ratio (defined as total adjusted capital divided by the authorized control level risk-based capital) generally exceeds 2. Therefore, when the authorized control level risk-based capital rises by \$1 million, the corresponding increase in surplus exceeds \$1 million. Conversely, column (2) of Table 3 shows that a \$1 million increase in risk-based capital raises prices by 0.095 percentage points. Taken together, the results in Table 3 highlight a trade-off between insurer solvency and policy affordability in

response to higher capital requirements.

Table 3: IV Regression

	<i>Dependent variable:</i>	
	surplus	price change
	(1)	(2)
required capital	3.098*** (0.688)	0.095*** (0.029)
Observations	16,655	16,655
R ²	0.400	0.0861

Note: *p<0.1; **p<0.05; ***p<0.01

Notes: This table shows regression results of instrumental variable regressions. The instrumental variable is treated $\times$ reform. The variable of interest is Risk-based capital. Except for price change, the unit of other variables is a million dollar. The unit of price change is percent.

Since insurers sometimes hold more capital than the total Risk-Based Capital (RBC) requirement, I conduct a robustness check using a subsample of insurers that are more tightly constrained by the regulation. Specifically, these are insurers whose average RBC ratios before the treatment are within one or two standard deviations of the capital requirement threshold. The results from these more binding samples are broadly consistent with those obtained from the full sample. Table 11 presents the first-stage regression results for the binding subsamples. Columns (1) to (3) report results for insurers whose average pre-treatment RBC ratios fall within one standard deviation of the regulatory threshold. Columns (4) to (6) show results for those within two standard deviations. The magnitudes of the coefficients are consistent with those from the full sample. For the subsample within one standard deviation, the reduced-form coefficients are larger than those in the full sample. This suggests the reform have larger impacts on marginal firms.

Table 12 shows the IV regression results for the binding subsamples. The magnitude of the coefficients is similar to those obtained from the full sample, reinforcing the robustness of the main findings. The IV regressions also show that the increase in capital requirement have

more prominent impacts on marginal firms. For insurers whose average pre-treatment RBC ratios fall within one standard deviation of the regulatory threshold, a \$1 million increase in capital requirements leads to a \$3.167 million increase in surplus, and raises prices by 0.116 percentage points.

Although some insurers may hold capital in excess of regulatory requirements, the stricter capital regulation still affects non-binding insurers. This is because the policy increases the probability of falling below the required threshold, thereby incentivizing insurers to hold additional capital as a buffer. The regulatory reform requires insurers to separately calculate required capital for hurricane and earthquake exposures, and to estimate future losses using scenario-based modeling rather than relying solely on historical losses. As a result, the regulation becomes more binding during periods of increased natural catastrophe activity. Since the policy change coincides with a period of heightened natural disasters, I am unable to fully disentangle the effects of the new capital regulation from those of the catastrophes themselves. However, this limitation is reasonable, as the policy is specifically designed to be most effective when natural catastrophes are more frequent.

4 Model

This section develops a simple theoretical model to illustrate the decisions of property insurers regarding capital, credit rating, pricing, and insolvency. The model incorporates limited liability, capital regulation, and the risk of catastrophe shocks.

4.1 Environment

This is a static model with two markets. The two markets are (1) states that are exposed to hurricane or earthquake risks, and (2) states that have less exposure to hurricane and earthquake. I use m to denote a market. Within each market, risk-averse households, indexed by i , demand property insurance. Property insurers, indexed by j , are risk-neutral

and compete by differentiating their products.

Two primary mechanisms monitor and promote insurer solvency. First, the Department of Insurance in each state regularly evaluates the capital adequacy of property insurers. There are two main types of capital requirements: minimum capital requirements and risk-based capital requirements. The minimum capital requirement specifies the minimum level of capital that property insurers are required to maintain to ensure their financial solvency. In contrast, risk-based capital (RBC) requirements determine the amount of capital an insurer should hold based on its underlying risk exposure. The required level of risk-based capital depends on factors such as the volume and composition of insurance business written, the use of reinsurance, and the insurer's holdings of risky assets. Second, credit rating agencies provide independent assessments of insurers' financial strength by assigning them credit ratings.

In the event of an insurer's failure, a state-level Guaranty Association, funded by assessments on the surviving insurers in that market, compensates the policyholders of the failed firm. Participation in this association is mandated by state law for all solvent insurers.

The model incorporates two sources of uncertainty: the realization of losses, L_{jm} , and the return on risky assets, r_2 . Loss rate $L_{jm} \in [0, \infty]$ realizes after solving optimization problems. Loss rates L_{jm} correlate within the same market. I assume the risky asset return r_2 is independent from the loss rate L_{jm} . Risky asset return r_2 can be negative, and has higher mean and variance than the safe asset return r_1 .

The following paragraph describes the timeline of the setup.

Timing:

Stage 1: Insurers set price and capital based on regulations.

Stage 2: Regulators assess capital adequacy. Rating agencies assign credit rating.

Stage 3: Consumers decide what insurance products to purchase.

Stage 4: Shocks realize. Some insurers may choose to go bankrupt. If insurers fail, State

Guaranty Funds charge taxes and bail out consumers.

4.2 Demand

The demand model is a simple logit model. The formula below shows the demand model.

$$u_{ijm} = \alpha_1 p_{jm} + \alpha_2 R_j + \xi_{jm} + \varepsilon_{ijm} \quad (3)$$

This formula indicates that consumer i 's utility from choosing insurance j in market m depends on the insurance price p_{jm} , the insurer's credit rating R_j , and unobserved product characteristics ξ_{jm} . Consumers care about credit ratings because they are not fully bailed out if the insurer becomes insolvent. The term ε_{ijm} represents idiosyncratic demand shocks, which are assumed to follow an i.i.d. Type I extreme value distribution.

4.3 Supply

Insurers know distribution of losses. However, the randomness of the model (risky asset return r_2 and loss rates L_{jm}) realizes after agents made decisions. Insurers choose pricing, capital, and reinsurance policies to maximize firm value relative to the opportunity cost of investing their assets in capital markets. Insurers are protected by limited liability and are subject to both minimum and risk-based capital requirements. In addition, rating agencies assign credit ratings that reflect insurers' financial strength. I also impose a cap on reinsurance, with the maximum amount of reinsurance specified as a function of the volume of insurance sold by the insurer. This restriction is intended to eliminate arbitrage opportunities arising from excessive reliance on reinsurance. We can write

$$\max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \mathbb{E}(\Pi_j + A_{1j} + A_{2j} - (1 + r_c)(A_{1j} + A_{2j})) \quad (4)$$

s.t.

$$\begin{aligned}
\Pi_j &= \max\left\{\left[\sum_m (p_{jm} - L_{jm} - Exp_j)q(\mathbf{p}_m, \mathbf{R}) - p^{re}Q_j^{re} - FC_j + r_1(A_{1j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{1j}}{A_{1j} + A_{2j}}) + \right. \right. \\
&\quad \left. \left. r_2(A_{2j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{2j}}{A_{1j} + A_{2j}}) + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d\}\}\right], \right. \\
&\quad \left. - (A_{1j} + A_{2j}) - \nu_j^d\right\} \quad \text{(Profits with limited liability)} \\
R_j &= g(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), Q_j^{re}, \frac{A_{2j}}{A_{1j} + A_{2j}}, A_{1j} + A_{2j}) \quad \text{(Credit rating)} \\
2 &\leq \frac{A_{1j} + A_{2j}}{\bar{K}(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), Q_j^{re}, A_{2j})} \quad \text{(Risk-based capital requirement)} \\
C_{\min} &\leq A_{1j} + A_{2j} \quad \text{(Minimum capital requirement)} \\
Q_j^{re} &\leq \alpha \sum_m q(\mathbf{p}_m, \mathbf{R}) \quad \text{(Cap of reinsurance)}
\end{aligned}$$

Insurers choose price p_{jm} , risk-free asset A_{1j} , risky asset A_{2j} , and reinsurance Q_j^{re} to maximize firm values. $\mathbf{p}_m$ is the price vector including p_{jm} and opponents' prices $p_{-j,m}$. Similarly, $\mathbf{R}$ is the credit rating vector including R_j and opponents' credit ratings R_{-j} . $q(\mathbf{p}_m, \mathbf{R})$ is the quantity of insurance, which is equal to premium collection divided by the unit price p_{jm} estimated in demand estimation. Notice that insurance quantity $q(\mathbf{p}_m, \mathbf{R})$ is a function of price vector $\mathbf{p}_m$ and credit rating vector $\mathbf{R}$ in the market. Similarly, the reinsurance quantity Q_j^{re} is defined as reinsurance premiums divided by reinsurance price index p^{re} .

Insurers compare their expected firm value with the value of their outside option to determine whether to remain in the property insurance market. The outside option is to invest in the capital market and earn an annual return of r_c . Introducing this outside option also prevents firms from optimally choosing an unbounded level of assets. Since this is a static model, the firm value is equal to the sum of profit Π_j and assets in hand. Insurers can choose to hold risk-free assets A_{1j} and risky assets A_{2j} . The profit is defined with limited liability. This means when the realized profit is lower than the loss of capital, the insurer

can choose to lose the capital. ν_j^d is a random shock to smooth the insolvency decision. The random shock of insolvency ν_j^d is an unobserved bankruptcy cost or franchise value. The higher the ν_j^d , the less likely that the insurer will choose to be insolvent. The profit without limited liability is the following expression.

$$\begin{aligned}
& \sum_m (p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R}) - p^{re} Q_j^{re} - FC_j + r_1(A_{1j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{1j}}{A_{1j} + A_{2j}}) + \\
& r_2(A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{2j}}{A_{1j} + A_{2j}}) + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm} q(\mathbf{p}_m, \mathbf{R})) - d\}\} \\
& = \sum_m (p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R}) - p^{re} Q_j^{re} - FC_j + r_1 A_{1j} + r_2 A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \cdot \frac{r_1 A_{1j} + r_2 A_{2j}}{A_{1j} + A_{2j}} \\
& + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm} q(\mathbf{p}_m, \mathbf{R})) - d\}\}
\end{aligned}$$

The profit without limited liability is calculated as total revenue minus claim amounts, operating expenses, payments to reinsurers, and fixed costs, while adding back investment income and recoveries from reinsurance.

One of the primary sources of income for a property insurer is the investment income generated from its insurance float. Insurers collect premiums upfront and subsequently use these funds to meet future claim payments. During the period between premium collection and claim settlement, the accumulated premium funds can be invested in financial assets, allowing insurers to earn investment returns before the corresponding claims become payable. In the model, I therefore incorporate the value of the insurance float into the insurer's profit function. I assume that premiums collected during the period can be invested throughout the period in both risk-free and risky assets. The insurer allocates these funds between the two assets according to its initial portfolio shares. This expression represents the insurer's float.

$$\sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \cdot \frac{r_1 A_{1j} + r_2 A_{2j}}{A_{1j} + A_{2j}}$$

The following extended expression represents the payment received from reinsurance.

$$-\frac{1}{k} \log \left(\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))} \right)$$

This expression is a smooth function of $\min\{C Q_j^{re}, \max\{0, \sum_m (L_{jm} q(\mathbf{p}_m, \mathbf{R})) - d\}\}$, a combination of softmin and softmax function. k is a smoothness parameter of the function. d is deductible of reinsurance. C is the multiplier of the reinsurance payment based on quantity of reinsurance Q_j^{re} .

Credit rating is equal to a $g(\cdot)$ function which depends on quantity of written premium in risky region $q(\mathbf{p}_1, \mathbf{R})$, quantity of written premium in less risky region $q(\mathbf{p}_2, \mathbf{R})$, quantity of reinsurance Q_j^{re} , share of risky assets $\frac{A_{2j}}{A_{1j} + A_{2j}}$, and total assets $A_{1j} + A_{2j}$. This definition of credit rating is a simplified version of how credit rating agency evaluate credit rating in reality.

Capital requirements mandate that the total capital on hand must be at least twice the authorized control level of Risk-Based Capital (RBC). If total capital falls below this threshold, insurers will need to submit an RBC corrective-action plan. Let $\tilde{K}(\cdot)$ denote the authorized control level RBC. The function $\tilde{K}(\cdot)$ depends on the quantity of written premium in the risky region, $q(\mathbf{p}_1, \mathbf{R})$, the quantity of written premium in the less risky region, $q(\mathbf{p}_2, \mathbf{R})$, quantity of reinsurance, Q_j^{re} , and risky assets, A_{2j} . This specification of the authorized control level RBC aligns with U.S. regulatory practice, with adjustments made to fit the model.

To compute the optimal choices of prices, assets, and reinsurance, I define a domain in which insurers remain solvent and transform the objective function into the following form.

$$\begin{aligned}
& \max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \iiint_{D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} [\sum_m ((p_{jm} - L_{jm} - Exp_j)q(\mathbf{p}_m, \mathbf{R})) - p^{re}Q_j^{re} \\
& - FC_j + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d_j\}\} + r_1(A_{1j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{1j}}{A_{1j} + A_{2j}}) + \\
& r_2(A_{2j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{2j}}{A_{1j} + A_{2j}})]f dV \\
& - \iiint_{\Omega/D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} (A_{1j} + A_{2j} + \nu_j^d)f dV - r_c(A_{1j} + A_{2j})
\end{aligned}$$

s.t.

$$R_j = g(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), Q_j^{re}, \frac{A_{2j}}{A_{1j} + A_{2j}}, A_{1j} + A_{2j}) \quad (\text{Credit rating})$$

$$2 \leq \frac{A_{1j} + A_{2j}}{\tilde{K}(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), Q_j^{re}, A_{2j})} \quad (\text{Risk-based capital requirement})$$

$$C_{\min} \leq A_{1j} + A_{2j} \quad (\text{Minimum capital requirement})$$

$$Q_j^{re} \leq \alpha \sum_m q(\mathbf{p}_m, \mathbf{R}) \quad (\text{Cap of reinsurance})$$

The blue expression is the profit when the insurer is solvent. The purple expression represents the lost capital when it becomes insolvent. I plug in the definition of credit rating and use implicit function theorem to find first-order conditions. Since sometimes the capital requirement may not be binding, I use Karush–Kuhn–Tucker conditions to find solutions to the optimization problem. For detailed derivation of first-order conditions and Karush–Kuhn–Tucker conditions, please refer to Appendix 8.4. Ω represents the full domain.

The domain $D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})$ where insurers are solvent is

$$\begin{aligned}
& [\sum_m (p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R}) - p^{re} Q_j^{re} - FC_j + r_1(A_{1j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{1j}}{A_{1j} + A_{2j}}) + \\
& r_2(A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{2j}}{A_{1j} + A_{2j}}) \\
& - \frac{1}{k} \log \left(\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))} \right)] \\
& \geq -(A_{1j} + A_{2j} + \nu_j^d)
\end{aligned}$$

$M = 2$ in this case because there are only 2 markets: (1) risky markets exposed to hurricane and earthquake, (2) other markets. The domain $D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})$ is defined as the region where the profits are higher than the loss of capital.

4.4 Welfare

I define welfare as the sum of consumer and producer surplus, net of insolvency costs.

$$\text{Welfare} = \text{Consumer Surplus} + \text{Producer Surplus} - \text{Insolvency Costs} \quad (5)$$

Producer surplus represents the total profit earned by producers in the market.

$$\text{Producer Surplus} = \sum_j \Pi_j$$

Consumer surplus is defined as the expected utility in dollar terms. The consumer surplus is normalized to outsider option j' . The following expression is the consumer welfare derived from social surplus function. M_m represents the market size in market m .

$$\text{Consumer Surplus} = - \sum_m [M_m \cdot \frac{1}{\alpha_1} \log(\sum_{k \in D} \exp(\alpha_1 p_{km} + \alpha_2 R_k + \xi_{km}) + cons)]$$

After normalize the utility of outside option to be 0, then the consumer surplus becomes

$$\text{Consumer Surplus} = - \sum_m [M_m \cdot \frac{1}{\alpha_1} \log(\sum_{k \in D} \exp(\alpha_1 p_{km} + \alpha_2 R_k + \xi_{km}) + 1)] \quad (6)$$

In the event of an insurer's failure, State Guaranty Funds provide a safety net for policyholders by covering their claims. To finance these obligations, these funds levy assessments on the remaining solvent insurers. This paper utilizes data on these guaranty association assessments to approximate the costs of insolvency, which are defined as the claims that insolvent firms are unable to fulfill. It is important to acknowledge that this is not a perfect measure of insolvency costs. The assessments are a transfer to policyholders and are subject to statutory coverage limits, meaning they may not fully capture the total losses incurred. Nevertheless, these assessment data represent the most suitable and currently available proxy for the direct costs of insurer insolvency.

4.5 Equilibrium

I focus on pure strategy subgame perfect equilibrium. The equilibrium is a set of households' insurance choices, insurers' decision $\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}$, such that the following statements hold:

1. Given price p_{jm} and rating R_j , households maximize utility by choosing insurance.

Consumers' discrete choice lead to logit demand market shares:

$$S_{jm} = \frac{\exp(\alpha_1 p_{jm} + \alpha_2 R_j + \xi_{jm})}{\sum_{k \in \mathcal{C}} \exp(\alpha_1 p_{km} + \alpha_2 R_k + \xi_{km})} \quad (7)$$

$\mathcal{C}$ is choice set.

2. Given the insurance demand function, $\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}$ solve insurers' optimization problem in equation (4).
3. Market clears: $q^d(\mathbf{p}_m, \mathbf{R}) = q^s(\mathbf{p}_m, \mathbf{R})$.
4. Assessments collected by Guaranty funds are equal to insolvency costs.

5 Estimation

This section describes the estimation methodology and presents the corresponding results for both the demand and supply sides of the model.

5.1 Demand Estimation

For demand estimation, I first construct moment conditions. The assumption of the moment condition is that the unobserved product characteristics are not correlated with the instrumental variable. Since I do not observe the uninsured, I choose an insurer j that has most observations as the outside option. Therefore, I revise the moment condition as

$$\mathbb{E}([\xi_{jmt} - \xi_{j'mt}][z_{jmt} - z_{j'mt}]) = 0$$

ξ_{jmt} is the unobserved product characteristics and z_{jmt} is the instrumental variable. Furthermore, since I only observe price change and I need to recover time 0 price from the demand estimation, to avoid multicollinearity, I take time difference to construct the moment condition. The final moment condition for demand estimation is

$$\mathbb{E}([(\xi_{jmt} - \xi_{jm,t-1}) - (\xi_{j'mt} - \xi_{j'm,t-1})][(z_{jmt} - z_{jm,t-1}) - (z_{j'mt} - z_{j'm,t-1})]) = 0$$

For instrumental variables, I use claim amount (losses) of the same insurer in another market as the instrument for price. Since insurers may adjust prices if there are losses from another market, the instrumental variable is relevant to prices. Besides, losses of property insurance from other markets are usually related to natural disasters or accidents, which are exogenous to prices. I use a credit rating standard change as the instrument for credit rating. The credit rating standard change is relevant because it will affect values of credit rating. Besides, the trigger of the standard change is a severe hurricane, which is exogenous. In 2005, credit rating standards became stricter after hurricane Katrina. For example, A.M. Best

imposed extra stress test for the 2nd following catastrophe event. When evaluate potential catastrophe losses, S&P changed from 100 year to 250 year modeling. Fitch changed from evaluation of single worst event to Tail Value at Risk (TVaR). The Tail at Value at Risk is to evaluate the average losses in worst-case scenarios. Combining TVaR and revision in catastrophe modeling, the insurer will need to hold 10% more capital in order to maintain the same Fitch credit rating, if the insurer had exposure to catastrophe risks. The instrument for credit rating is a difference-in-difference type of instrumental variable. The instrument is $\text{treated}_j \times \text{reform}_t$. The variable reform_t is equal to 1 after year 2006 when the change of credit rating standards happened. Treated_j is defined as the following equation.

$$\text{treated}_j = \frac{\text{property underwriting in treated states}_j}{\text{total underwriting}_j}$$

From the definition above, treated_j is defined as the share of property underwriting in treated states as of the total underwriting. If an insurer has more property underwriting in the disaster-prone regions, then this insurer is more likely to be treated.

Since I only observe price change and revenue market share data, after some algebra, the difference in unobserved product characteristics is defined as

$$\begin{aligned} & (\xi_{jmt} - \xi_{jm,t-1}) - (\xi_{j'mt} - \xi_{j'm,t-1}) \\ &= (\log \omega_{jmt} - \log \omega_{jm,t-1}) - (\log \omega_{j'mt} - \log \omega_{j'm,t-1}) - \tilde{\alpha}_{1jm}(d_{jmt} - d_{jm,t-1}) \\ & \quad + \tilde{\alpha}_{1j'm}(d_{j'mt} - d_{j'm,t-1}) - (\log d_{jmt} - \log d_{jm,t-1}) + (\log d_{j'mt} - \log d_{j'm,t-1}) \\ & \quad - \alpha_2[(R_{jt} - R_{j,t-1}) - (R_{j't} - R_{j',t-1})] \end{aligned}$$

For detailed derivation of the above expression, please see Appendix 8.5. In the equation above,

$$d_{jmt} = \prod_{s=1}^t \frac{p_{jms}}{p_{jm,s-1}}$$

This means d_{jmt} is the cumulative price change across years.

$$\tilde{\alpha}_{1jm} = \alpha_1 p_{jm0},$$

$$\tilde{\alpha}_{1j'm} = \alpha_1 p_{j'm0}$$

With this method, I can directly estimate $\tilde{\alpha}_{1jm}$ and $\tilde{\alpha}_{1j'm}$. Suppose there are 1000 combination of j and m . Since α_1 is common across all $\tilde{\alpha}_{1jm}$ and $\tilde{\alpha}_{1j'm}$, there will be 1000 equations and 1001 unknowns. Therefore, I normalize the time 0 price of the median firm to be 1 and solve for the rest of the α_1 , p_{jm0} , and $p_{j'm0}$.

However, estimating a unique baseline price p_{jm0} for every individual insurer would create data sparsity issues. To pool observations and obtain sufficient statistical power, I group insurers into $G = 5$ size groups (quintiles). To ensure that group assignment is strictly predetermined and exogenous to subsequent market demand shocks, an insurer's group $g(j) \in \{1, 2, 3, 4, 5\}$ is assigned based on its initial firm size—defined as its total underwriting volume in the very first year firm j appears in the dataset. For firms that enter the market later, I adjust d_{jmt} using the average market price change to align their prices with the initial year of the group. All firms within the same group will share the same initial price. Grouping by initial underwriting size captures structural differences in insurer scale, which may impact the initial price p_{jm0} .

The estimated coefficients are $\hat{\alpha}_1 = -1.0157$ and $\hat{\alpha}_2 = 0.1173$. Table 4 displays the results. Price elasticity is computed using the following expression:

$$\text{price elasticity}_{jm} = \alpha_1 \times p_{jm} \times (1 - S_{jm})$$

where p_{jm} denotes the price and S_{jm} represents the market share. The median estimated price elasticity is -1.012 . The scale of the price elasticity is similar to the price elasticity estimated in Grace et al. (2004), which find that the elasticity for catastrophe coverage is

around -1.9 and the elasticity for non-catastrophe coverage is around -0.4. Figure 4 presents the distributions of the estimated time-0 prices and price elasticities.

Table 4: Estimation Results of Demand Parameters

Variable	Coefficient
α_1 (Price Sensitivity)	-1.0157*** (0.0011)
α_2 (Credit Rating)	0.1173*** (0.0149)

Note: Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

This table shows estimation results and standard errors of demand estimation.

5.2 Supply Estimation

On the supply side, I estimate the key components required for the counterfactual analysis. The random shocks ν_j^d are inferred based on the Karush–Kuhn–Tucker (KKT) conditions and insurers’ insolvency decisions, following an approach similar to the supply-side estimation in BLP. The underlying assumption is that the observed data reflect the optimal decisions of insurers in a competitive market. The quantity of reinsurance, Q_j^{re} , is computed by dividing the observed reinsurance premium by an approximate unit price p^{re} . Note that the unit price p^{re} is assumed to be identical across all insurers. I estimate the parameters of the required capital function $\tilde{K}(\cdot)$ and the credit rating function $g(\cdot)$ using linear functions. Parameters governing the distributions of losses and returns on risky assets are also estimated from the data. In addition, I estimate the functions determining reinsurance payments and the cap of reinsurance holding directly from observed data. Finally, I use assessments from the Guaranty Association to approximate the social costs associated with insurer insolvency.

Table 5: Supply Estimation

	Parameters or variables	Method
ν_j^d	Random shocks in insolvency decisions	Estimation of supply system (FOC + insolvency decisions)
Q_j^{re}	Reinsurance quantity	Calculate from reinsurance premiums, by dividing p^{re}
L_{jm}, r_2	Parameters of shocks	Estimate from data
$\tilde{K}(\cdot), g(\cdot)$	Policy and credit rating functions	Estimate from data
C, d, k	Parameters of payments of reinsurance	Estimate from data
α	Parameters of cap of reinsurance	Estimate from data
	Insolvency social costs	Estimate from Guaranty association assessments

Notes: This table shows items that need to be estimated in the supply estimation. The second column shows the parameters and variables to be estimated. The last column explains methods to estimate parameters.

5.2.1 Required Capital Function

I use linear models to estimate required capital function $\tilde{K}(\cdot)$. Please see the expression below.

$$\tilde{K}(X_1, X_2, X_3) = \sum_{l=1}^3 \beta_l X_l$$

Inputs in the linear models are the same as the $\tilde{K}(\cdot)$ function. These variables are written premium in risky market, written premium in less risky market, reinsurance, and risky assets. Please see the required capital function below.

$$\tilde{K}(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), A_{2j})$$

Since there was a policy change in 2017, I separately estimate the required capital function for the periods before and after the policy change. The out-of-sample R^2 of the test sample is 88.11% before the policy change and 87.57% afterward. Table 14 presents the estimation results of the required capital function using linear models. The results indicate a positive relationship between underwriting in risky markets and the required capital, as well as a positive correlation between the level of risky assets and the required capital, suggesting that firms with higher exposure to risky markets and risky assets are required to hold more capital.

5.2.2 Credit Rating Function

Similar to required capital function, I use the following linear model to estimate the credit rating function $g(\cdot)$.

$$g(X_1, X_2, \dots, X_4) = \sum_{l=1}^4 \beta_l X_l$$

Variables X_1 to X_4 correspond to variables in the credit rating function $g(\cdot)$. These variables are written premium in risky market, written premium in less risky market, reinsurance, share of risky assets, and total assets. Please see the credit rating function below.

$$R_j = g(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), \frac{A_{2j}}{A_{1j} + A_{2j}}, A_{1j} + A_{2j})$$

Since the credit rating standards changed in 2005, I estimate the credit rating function separately for the periods before and after the policy change. The out-of-sample accuracy rate from the estimation is approximately 0.488 before the policy change and 0.48 after it. Although these accuracy rates are not particularly high, they represent the best predictive performance obtained in my analysis. Table 15 illustrates the relationship between the independent variables and credit ratings prior to the standard change. The results suggest that larger total assets, higher reinsurance, and higher underwriting in less risky markets are associated with higher credit ratings, while a higher share of risky assets is associated with lower credit ratings.

5.2.3 Distributions, Payment from Reinsurance, Cap of Reinsurance, and Insolvency Costs

I estimate the distribution of returns on risky assets and loss rates from the data. The normal inverse Gaussian (NIG) distribution, which has a closed-form probability density function (pdf), fits the empirical distribution of returns on risky assets well. See Figure 6 for the estimated distribution. For detailed parameter estimates and the pdf of the NIG distribution, please refer to Appendix 8.8.

Due to the high frequency of zero-valued observations in loss rates, I estimate their distribution using a hurdle log-normal model. The estimated distributions are displayed in Figure 7. The results show that the mean loss rate in risky markets is higher than in less risky markets. I define risky markets as those with exposure to hurricane and earthquake risks. Details on parameter estimation and the pdf of the hurdle log-normal distribution can be found in Appendix 8.9.

The reinsurance payment function is also estimated from the data. The original payment function is defined as:

$$\min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d\}\}$$

To facilitate differentiation during the optimization process, I approximate the max and min functions using smooth counterparts: softmax and softmin. The smoothed reinsurance payment function becomes:

$$-\frac{1}{k} \log \left(\exp(-k \cdot CQ_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm}q(\mathbf{p}_m, \mathbf{R}) - d))} \right)$$

For a detailed derivation of this approximation, please refer to Appendix 8.10. The reinsurance deductible d and the smoothness parameter k are estimated from the data. The results suggest that $k = 5$, $C = 1.14$ and $d = 0$.

Regarding the cap on insurance holdings,

$$Q_j^{re} \leq \alpha \sum_m q(\mathbf{p}_m, \mathbf{R})$$

I approximate α empirically. Specifically, I calculate the ratio of $\sum_m q(\mathbf{p}_m, \mathbf{R})$ to Q_j^{re} for each observation. I then set α equal to the 99th percentile of the resulting distribution of ratios. The estimated α is 4.18.

In addition, the minimum capital requirement, C_{min} , is set at \$2 million, which is broadly consistent with the average minimum capital requirement for property insurers in the United States. The annual return on the outside option, r_c , is calibrated to 6.5%, which is comparable to the historical real rate of return on capital market investments in the United States after accounting for inflation.

Finally, I measure the social cost of insolvency using Guaranty Association assessments. These represent the claims left unpaid by insolvent insurers and subsequently borne by the public. On average, each insolvency leads to a societal loss of approximately \$25.095 million.

5.2.4 Random Shocks of Insolvency

During the supply-side estimation, most parameters are identified directly from the data. To estimate the random shocks of insolvency, I rely on a revealed preference approach, assuming that the observed choices represent insurers' optimal decisions in the market. I then recover the insolvency shocks ν_j^d by solving for values that satisfy the Karush–Kuhn–Tucker (KKT) conditions. This procedure allows ν_j^d to capture the discrepancy between the theoretical model and real-world behavior, effectively absorbing model misspecification or unobserved heterogeneity.

To structurally estimate the unobserved random shocks of insolvency, ν_j^d , I invert the insurers' optimal supply-side first-order conditions (FOCs). Following the BLP literature framework (e.g., Berry et al. (1995)), I assume that the observed prices, asset allocations,

and reinsurance choices of the insurers represent their ex-ante optimal decisions.

Let $\mathbf{V} = (L_{j1}, L_{j2}, r_2)$ denote the vector of random state variables (catastrophe losses in both markets and the risky asset return) drawn from the joint density $f(\mathbf{V})$. I define the pre-default terminal wealth of firm j as:

$$W_j(\mathbf{V}; \mathbf{x}_j) = \sum_{m=1}^2 (p_{jm} - L_{jm} - \text{Exp}_j) q_{jm} - p^{re} Q_j^{re} - \text{FC}_j + \text{ReinsPayout}(\mathbf{V}) + (1 + r_1) A_{1j} + (1 + r_2) A_{2j} + \text{float}(\mathbf{V}) \quad (8)$$

where $\mathbf{x}_j = (p_{j1}, p_{j2}, A_{1j}, A_{2j}, Q_j^{re})$ is the vector of choice variables.

Due to limited liability, the firm defaults if its terminal wealth falls below the threshold of its unobserved franchise value, $\nu_j^d \geq 0$. The firm's expected objective function is:

$$\max_{\mathbf{x}_j} \mathbb{E}[\pi_j] = \iiint_{\mathbf{V} \in D_j} (W_j(\mathbf{V}; \mathbf{x}_j) + \nu_j^d) f(\mathbf{V}) d\mathbf{V} - \nu_j^d - (1 + r_c)(A_{1j} + A_{2j}) \quad (9)$$

where $D_j = \{\mathbf{V} \mid W_j(\mathbf{V}; \mathbf{x}_j) \geq -\nu_j^d\}$ defines the solvency domain.

The firm maximizes its expected payoff subject to two primary regulatory constraints: the risk-based capital (RBC) requirement and the absolute minimum capital requirement, assuming the cap of reinsurance holding constraint is not binding, which is the case for majority firms in the data. The Lagrangian is:

$$\mathcal{L} = \mathbb{E}[\pi_j] + \lambda_j (A_{1j} + A_{2j} - 2\tilde{K}(\mathbf{x}_j)) + \mu_j (A_{1j} + A_{2j} - C_{\min}) \quad (10)$$

By differentiating the Lagrangian with respect to any decision variable $x \in \mathbf{x}_j$, the ex-ante First-Order Condition is:

$$\mathcal{F}_x(\mathbf{x}_j; \nu_j^d, \lambda_j, \mu_j) \equiv \iiint_{D_j} \frac{\partial W_j(\mathbf{V})}{\partial x} f(\mathbf{V}) d\mathbf{V} - \text{MC}(x) = 0 \quad (11)$$

where $\text{MC}(x)$ represents the marginal cost of the decision (e.g., the opportunity cost of capital $1 + r_c$ combined with the shadow costs of the regulatory constraints $\lambda_j \nabla_x \tilde{K}$ and μ_j). Note that because the boundary of integration D_j is defined by $W_j(\mathbf{V}) = -\nu_j^d$, the boundary terms in the derivative evaluate to zero. Appendix 8.4 shows the detailed expressions and derivation of the first-order conditions.

I observe the firm's actual choices $\mathbf{x}_j^{\text{obs}}$ and determine empirically which constraints are binding in the data. I partition the estimation into four Karush-Kuhn-Tucker (KKT) regimes:

1. Unconstrained: ($\lambda_j = 0, \mu_j = 0$). I solve for ν_j^d .
2. Only RBC constraint binds: ($\lambda_j > 0, \mu_j = 0$). I jointly estimate (ν_j^d, λ_j) .
3. Only minimum capital binds: ($\lambda_j = 0, \mu_j > 0$). I jointly estimate (ν_j^d, μ_j) .
4. Both bind: ($\lambda_j > 0, \mu_j > 0$). I jointly estimate $(\nu_j^d, \lambda_j, \mu_j)$.

A critical feature of this structural model is that ν_j^d is point-identified for marginally solvent and insolvent firms, but it is only set-identified (bounded) for deeply solvent firms.

To understand why, let's examine the sensitivity of the FOC in Equation 11 to the unobserved parameter ν_j^d . By applying the Multivariate Leibniz Integral Rule, the derivative of the FOC with respect to ν_j^d is a surface integral over the default boundary ∂D_j :

$$\frac{\partial \mathcal{F}_x(\mathbf{x}_j; \nu_j^d)}{\partial \nu_j^d} = \iint_{W_j(\mathbf{V}) = -\nu_j^d} \nabla_x W_j(\mathbf{V}) \cdot \frac{f(\mathbf{V})}{\|\nabla_{\mathbf{V}} W_j(\mathbf{V})\|} dS \quad (12)$$

The derivation of Equation 12 can be found in Appendix 8.6. This gradient dictates whether the model can uniquely invert $\mathbf{x}_j^{\text{obs}}$ to find ν_j^d .

For marginal solvent firms, the observed choices of thin capital and low pricing place the default threshold $W_j(\mathbf{V}) = -\nu_j^d$ inside a plausible region of the loss distribution (e.g., a 1-in-20 year catastrophe). Therefore, the probability density on the boundary is strictly

positive $(f(\mathbf{V})|_{W_j(\mathbf{V})=-\nu_j^d} > 0)$. Furthermore, in catastrophe states, the marginal payoff of choices like price or reinsurance is strongly non-zero ($\nabla_x W_j \neq 0$). Proof of $\nabla_x W_j \neq 0$ is in Appendix 8.7. Consequently, $\frac{\partial \mathcal{F}}{\partial \nu_j^d} \neq 0$. The FOC is strictly monotonically increasing or decreasing in ν_j^d , guaranteeing a unique, point-identification for the random shocks of insolvency ν_j^d .

For highly capitalized firms (deeply solvent), default only occurs under impossibly extreme theoretical events. The boundary $W_j(\mathbf{V}) = -\nu_j^d$ lies in the deep tail where the probability density is effectively zero ($f(\mathbf{V}) \approx 0$). As a result, the surface integral in Equation 12 collapses to zero. Changing ν_j^d does not meaningfully alter the firm's expected risk exposure. The FOC gradient with respect to ν_j^d is flat, meaning the parameter is set-identified $[0, \infty)$. The random shocks of insolvency ν_j^d is positive because it represents the extra franchise values that the firm will lose if it defaults. When $\nu_j^d < 0$, there would be arbitrage opportunity, which means firms might choose to enter the market with minimum capital and default to "earn" the $-\nu_j^d$ value, if $-\nu_j^d - A_{1j} - A_{2j} > A_{1j} + A_{2j}$. Therefore, ν_j^d is positive to prevent arbitrage.

To structurally estimate the parameters, I define the objective function as the sum of squared FOC residuals normalized by the probability of the solvency domain, subject to the economic restriction that franchise value must be non-negative ($\nu_j^d \geq 0$). As mentioned in the previous paragraph, negative values of ν_j^d are economically irrational as they would imply a firm receives a cash subsidy upon declaring bankruptcy.

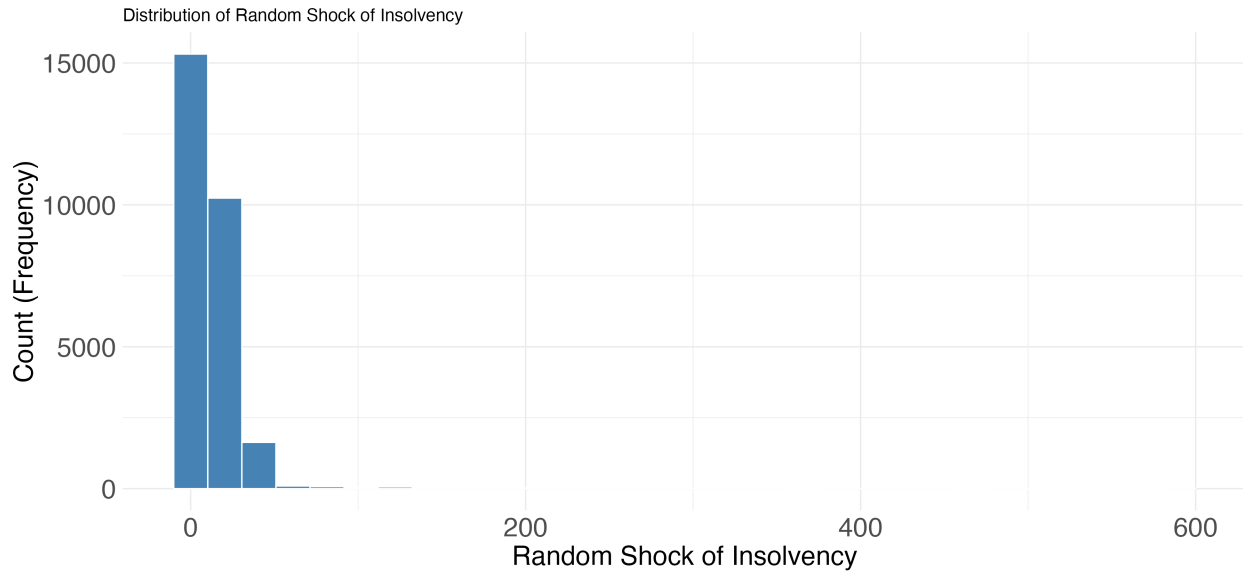
I normalize the sum of squared first-order condition residuals by the solvency probability $\mathbb{P}(\mathbf{V} \in D_j)$ to ensure both numerical stability and scale invariance. First, because $\lim_{\mathbb{P}(D_j) \rightarrow 0} \iiint_{D_j} \nabla_x W_j f(\mathbf{V}) d\mathbf{V} = 0$, dividing by $\mathbb{P}(\mathbf{V} \in D_j)$ creates an asymptotic barrier ($\frac{1}{\mathbb{P}(D_j)} \rightarrow \infty$) that prevents the optimizer from collapsing the integration domain to zero to spuriously minimize residuals. Second, by rewriting the integral via conditional expectation as $\mathbb{P}(\mathbf{V} \in D_j) \mathbb{E}[\nabla_x W_j | \mathbf{V} \in D_j]$, this normalization evaluates errors relative to per-state

optimality conditional on survival, preventing the solver from artificially underweighting high-risk firms with smaller solvency masses.

Because the model only point-identifies ν_j^d for marginally solvent and insolvent firms, relying on the point estimates of deeply solvent firms ($\hat{\nu}_j^d \approx 0$) would misrepresent their true unobserved franchise value in a counterfactual scenario where they might choose to take on more risk. To address this, I only use the estimation results where $\hat{\nu}_j^d > 0.1$ to fit the parameters of the underlying franchise value distribution. In subsequent counterfactual analyses, the unobserved franchise values for all firms are drawn from this strictly positive fitted distribution.

Figure 3 presents the estimated distribution of ν_j^d . The median value is \$8.7 million, and the mean value is \$15.08 million. Furthermore, the third-quartile value is \$18.06 million. Thus, although the distribution exhibits a long right tail, the tail is relatively thin, indicating that observations in the upper tail are comparatively sparse.

Figure 3: Estimation Results of Random Shocks of Insolvency



Notes: This figure shows results of estimation of random shocks of insolvency ν_j^d . The unit is a million dollars.

6 Counterfactual Policy Analysis

In this section, I conduct counterfactual analyses to evaluate the trade-off between insurer solvency and policy affordability as capital regulation becomes stricter. My theoretical framework allows for a deeper exploration of this trade-off than is possible with reduced-form methods alone. While the reduced-form evidence presented in Section 3 establishes the statistical relationships between capital regulation, insolvency risk, and premiums, it cannot fully capture the underlying economic mechanisms driving these outcomes or predict how insurers and consumers would behave under policy regimes that have not been observed. Second, the structural model enables me to quantify the welfare trade-offs at play. For instance, I can simulate the impact of a heightened regulatory threshold on the equilibrium, observing how it affects insurer failure rates (solvency) and the resulting premium adjustments (affordability). This allows for a quantitative assessment of the costs and benefits of such a policy change from the perspectives of both consumers and insurers. More importantly, counterfactual analyses enable us to examine scenarios that are not observed in the data, providing valuable insights into the potential effects of alternative policies and informing policy decisions.

The initial counterfactual analysis will focus on tightening the existing capital requirements. This exercise will trace out the efficiency frontier between solvency and affordability, providing a nuanced understanding of the potential consequences of various policy choices. Building on these insights, the second part of the counterfactual analysis will then disentangle the roles of capital regulation and credit ratings in mitigating insolvency risk. By simulating a scenario where capital regulation is removed, I can evaluate the extent to which market-based discipline, through credit ratings, can effectively constrain insurer risk-taking. This will shed light on the unique contribution of formal regulation in the insurance market. In the third part of the counterfactual analysis, I consider a scenario in which there is no capital requirement, while consumers place a high weight on credit ratings when making financial

decisions. This counterfactual examines whether credit ratings can serve as a substitute for capital requirements when consumers are highly attentive to insurers' creditworthiness. Since holding capital is costly for financial institutions, the analysis asks whether greater financial literacy among consumers—and the resulting increase in their ability to assess and respond to credit ratings—could provide sufficient market discipline to mitigate the need for regulatory capital requirements.

In the counterfactual analysis, insurers choose price, asset allocation, and reinsurance to maximize expected firm values under limited liability. During the optimization process, since the choice of price p_{jm} and the opponents' prices $p_{-j,m}$ jointly determine the quantity $q(\mathbf{p}_m, \mathbf{R})$, all insurers make their decisions simultaneously each year in the algorithm. If no firm has an incentive to deviate from its chosen price, asset allocation, or reinsurance strategy, the outcome converges to a Nash equilibrium. The quantity of insurance is given by

$$q(\mathbf{p}_m, \mathbf{R}) = \frac{e^{\delta_{jm}}}{\sum_k e^{\delta_{km}}} \cdot N_m$$

where $\delta_{jm} = \alpha_1 p_{jm} + \alpha_2 R_j + \xi_{jm}$. N_m is the market size which is the total quantity of insurance in a market m .

I made two adjustments in the counterfactual analyses to bridge the reduced-form estimates of the model with real-world economic constraints, explicitly addressing the Lucas Critique [Lucas (1976)].

First, as Table 15 suggests, the estimated coefficients for the credit rating function are relatively small because all firms in the sample operated under a regime where they were legally required to hold at least a statutory minimum capital ($C_{\min}$) historically. This means historically the scale of asset holding is relatively large. If capital requirements are removed in a counterfactual scenario, the small coefficients would cause a failure of the Lucas Critique: the optimizer would assume a firm holding zero assets could still receive an A-grade rating simply due to the statistical intercept. To resolve this, I introduce a solvency anchor that

scales the predicted rating proportionally if a firm’s total capital falls below the statutory minimum threshold ($R_{\text{adj}} = R_{\text{calc}} \times \min(1, A_{\text{tot}}/C_{\text{min}})$). This would enable the usage of the coefficients in Table 15 more reasonable in counterfactual analyses.

Second, I impose a structural constraint on the relationship between reinsurance and credit ratings. A linear regression mathematically treats reinsurance (Q_j^{re}) and capital ($A_{1j} + A_{2j}$) as perfect substitutes. Without adjustment, the numerical solver could exploit this linearity by allowing a firm to hold zero capital and theoretically buy a AAA rating entirely through reinsurance contracts. In reality, rating agencies view capital and reinsurance as structural complements because reinsurers are also subject to credit risks. To reflect this, the model caps the effective reinsurance recognized by the rating agency at a realistic multiple of the firm’s actual capital base ($Q_{\text{effective}}^{re} = \min(Q_j^{re}, 10 \times A_{\text{tot}})$). This prevents firms from utilizing the linear econometric specification as a loophole to bypass capital accumulation.

Unlike the credit rating function, the estimated capital requirement function, $\tilde{K}(\cdot)$, does not require any scaling adjustments because it is fundamentally immune to the Lucas Critique in this context. This distinction relies on economic and mathematical factors: First, capital regulation $\tilde{K}$ is in the denominator, and the estimated $\tilde{K}$ equation possesses a zero intercept and strictly positive coefficients. Consequently, there are no loopholes or negative weights that an optimizer can exploit to artificially suppress its capital requirement. Second, the capital regulation $\tilde{K}$ only shows up in the constraint of the optimization problem. The credit rating function, however, would enter into the demand function, and creates feedback effects. With these two reasons, the capital requirement function is immune to the Lucas Critique, while the credit rating function is not.

6.1 A World with Stringent Capital Requirement

Building on the baseline empirical results, this counterfactual analysis explores the mechanism of capital regulation by evaluating its impact on the trade-off between solvency and affordability. I simulate market outcomes under progressively stricter capital requirement

thresholds, κ . Within the model, all agents optimize given the regulatory environment, as consumers maximize utility and insurers maximize firm values. The objective is not to identify a single optimal threshold, κ^* , but rather to characterize how equilibrium solvency and affordability levels move in response to changes in κ . This exercise provides direct insight into the costs and benefits of tightening capital regulation.

$$\kappa \leq \frac{A_{1j} + A_{2j}}{\bar{K}(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), Q_j^{re}, A_{2j})} \quad (\text{Risk-based capital requirement})$$

Recall that welfare is

$$\text{Welfare} = \text{Consumer Surplus} + \text{Producer Surplus} - \text{Insolvency Costs}$$

Because of computational constraints, I evaluate only several threshold levels and compare the resulting welfare outcomes with those from the baseline setup. In the baseline setup, the risk-based capital requirement threshold is set at 2, consistent with the threshold imposed in the real-world setup. In the following table, the capital threshold is set to $\kappa = 6$. At this threshold, approximately 25% of property insurers in the original sample hold capital below this level. After tightening the capital regulation, both solvency and insurance prices increase. The insolvency rate decreases by 0.14 percentage point. Price in risky market increases by 3.9%. This result is consistent with the reduced-form evidence in Section 3, which indicates capital regulation trades off solvency and affordability. In this counterfactual analysis, the higher prices reduce insurance demand, and operation profits of insurers reduce by 5.45% (\$2.49 million) on average. In order to comply with the stringent risk-based capital requirement, insurers hold more risk-free and risky assets. The asset holding increases by 178.8% (\$20.54 million) on average. Since they hold more assets, the investment profits also increase. The Herfindahl-Hirschman Index also increases, which means markets become more concentrated. In this case, consumer surplus decreases because consumers face

a smaller variety of products in the market. Since profits reduce, producer surplus decreases. Although a stringent risk-based capital requirement reduces the society's insolvency costs, the welfare gains from lower insolvency risk are outweighed by the associated reductions in consumer and producer surplus. Because insolvency is a relatively infrequent event, its costs are smaller in magnitude than the changes in consumer and producer surplus. Consequently, aggregate welfare under a stringent risk-based capital requirement is lower than in the baseline setup.

The definition of insolvency costs in Table 6 is number of insolvency times average costs of insolvency per event. In Columns (3) and (4) in Table 18, I calculate the second definition of insolvency costs. In the second definition, the insolvency cost is equal to the realized losses minus total assets, shown in Equation 13.

$$\begin{aligned}
& - [\sum_m (p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R}) - p^{re} Q_j^{re} - FC_j + r_1 A_{1j} + r_2 A_{2j} + \\
& \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \cdot \frac{r_1 A_{1j} + r_2 A_{2j}}{A_{1j} + A_{2j}} + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm} q(\mathbf{p}_m, \mathbf{R})) - d\}\} + A_{1j} + A_{2j}]
\end{aligned} \tag{13}$$

The insolvency costs calculated under the second definition are higher than those calculated under the first definition. Nevertheless, the magnitude of the insolvency costs remains smaller than that of the consumer and producer surpluses. Consequently, aggregate welfare under the stringent capital requirement (row 2) is lower than in the baseline setup (row 1). Table 18 also reports the insolvency costs and aggregate welfare for the other counterfactual analyses. These counterfactual analyses are discussed in the following subsections.

Table 6: Higher threshold of capital requirement decreases insolvency and affordability

Variable	Mean			Median		
	Baseline	Counterfactual	Diff	Baseline	Counterfactual	Diff
	(1)	(2)	(3)	(4)	(5)	(6)
insolvency	0.0067	0.0052	-0.0014	0.0000	0.0000	0.0000
p in risky market	1.9082	1.9832	0.0750	1.8423	1.9283	0.0860
p in less risky market	2.0962	2.6195	0.5233	1.8909	2.0194	0.1286
q in risky market	27.7211	24.8034	-2.9177	6.7679	5.8233	-0.9446
q in less risky market	9.2884	8.7051	-0.5833	3.0043	2.4185	-0.5858
profits	49.0937	46.6024	-2.4913	18.8682	16.8688	-1.9994
operation profits	44.5993	42.1665	-2.4328	18.1630	16.0319	-2.1311
investment profits	2.1994	2.6858	0.4864	0.6268	0.6712	0.0444
risky assets	5.2746	17.2325	11.9580	0.0297	1.8894	1.8597
total assets	11.4897	32.0346	20.5448	2.0013	3.3335	1.3322
Q_{re}	5.8704	5.7053	-0.1651	0.0023	0.0165	0.0143
HHI in risky market	0.0115	0.0118	0.0003	0.0105	0.0106	0.0001
HHI in less risky market	0.0101	0.0103	0.0003	0.0097	0.0104	0.0007
consumer surplus	585848.3701	581863.0784	-3985.2917	608838.7637	603565.5160	-5273.2477
producer surplus	37596.6521	35607.6489	-1989.0032	36699.9523	35264.3778	-1435.5746
insolvency cost	128.2633	100.3800	-27.8833	125.4750	100.3800	-25.0950
welfare	623316.7589	617370.3473	-5946.4116	648808.4123	640517.3400	-8291.0723

Notes: This table shows comparison of statistics of original data and counterfactual. The first three columns display mean of variables. The last three columns display median of variables. Columns (1) and (4) show statistics of original data. Columns (2) and (5) show results of counterfactual analysis when threshold of capital requirement is high. Column (3) is equal to column (2) minus column (1). Similarly, column (6) is equal to column (5) minus column (4). The unit of q and Q_{re} is a million. Except for insolvency, p, q, Q_{re} , and HHI, the unit of variables is a million dollars.

6.2 A World without Capital Regulation

The second counterfactual analysis disentangles the respective roles of capital regulation and credit ratings in constraining insolvency risk. I examine whether credit ratings alone can effectively prevent insolvency. To isolate the role of ratings, I remove risk-based capital regulation from the optimization problem. The resulting optimization problem is given by

$$\begin{aligned}
& \max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \mathbb{E}(\Pi_j + A_{1j} + A_{2j} - (1 + r_c)(A_{1j} + A_{2j})) \\
& s.t. \\
& \Pi_j = \max\{[\sum_m (p_{jm} - L_{jm} - Exp_j)q(\mathbf{p}_m, \mathbf{R}) - p^{re}Q_j^{re} - FC_j + r_1(A_{1j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{1j}}{A_{1j} + A_{2j}}) + \\
& \quad r_2(A_{2j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{2j}}{A_{1j} + A_{2j}}) + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d\}\}], \\
& \quad - (A_{1j} + A_{2j}) - \nu_j^d\} \quad \text{(Profits with limited liability)} \\
& R_j = g(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), Q_j^{re}, \frac{A_{2j}}{A_{1j} + A_{2j}}, A_{1j} + A_{2j}) \quad \text{(Credit rating)} \\
& C_{\min} \leq A_{1j} + A_{2j} \quad \text{(Minimum capital requirement)} \\
& Q_j^{re} \leq \alpha \sum_m q(\mathbf{p}_m, \mathbf{R}) \quad \text{(Cap of reinsurance)}
\end{aligned}$$

Removing the risk-based capital requirement would bring the market closer to the institutional environment that prevailed before the 1990s, when U.S. property and casualty insurers were subject primarily to fixed minimum capital requirements. As documented in Appendix 8.3, several P&C insurers failed during the 1980s after aggressively expanding their business to capitalize on high interest rates. The fixed capital requirement was insufficient to constrain such rapid growth. When interest rates subsequently declined and natural disasters generated substantial underwriting losses, a number of insurers became insolvent.

Table 7 reports results that are consistent with this historical experience. Eliminating the risk-based capital requirement increases the insolvency rate by 0.07 percentage points. At the same time, insurance becomes more affordable, with prices declining by approximately

1.2% in the risky market. The resulting increase in demand leads to higher insurance sales and operating profits, which rise by 15.9%, or \$7.1 million. Insurers also reduce their holdings of both safe and risky assets. Average total asset holdings decline to \$2.33 million, only slightly above the minimum capital requirement. As a result, investment profits fall by 18%, or \$0.398 million. Insurers also reduce their reinsurance holdings.

The lower capital requirement reduces the barrier to entry, leading to a decline in the Herfindahl–Hirschman Index (HHI) in both markets and, consequently, greater market competition. Increased competition lowers insurance prices and raises consumer surplus. At the same time, the increase in operating profits raises producer surplus. These gains are accompanied by a 12.31% increase in insolvency costs, equivalent to \$15.8 million. Under the second definition of insolvency costs, the total cost is \$2,026.07 million, as reported in Table 18. Nevertheless, aggregate welfare increases because the gains in consumer and producer surplus more than offset the increase in insolvency costs.

In robustness check, I also experiment with removing both risk-based capital requirement and minimum capital regulation. In this case, the optimization problem becomes

$$\begin{aligned}
& \max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \mathbb{E}(\Pi_j + A_{1j} + A_{2j} - (1 + r_c)(A_{1j} + A_{2j})) \\
& s.t. \\
& \Pi_j = \max\left\{\left[\sum_m (p_{jm} - L_{jm} - Exp_j)q(\mathbf{p}_m, \mathbf{R}) - p^{re}Q_j^{re} - FC_j + r_1(A_{1j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{1j}}{A_{1j} + A_{2j}}) + \right. \right. \\
& \quad \left. r_2(A_{2j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R})\frac{A_{2j}}{A_{1j} + A_{2j}}) + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d\}\}\right], \\
& \quad \left. - (A_{1j} + A_{2j}) - \nu_j^d\right\} \quad \text{(Profits with limited liability)} \\
& R_j = g(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), Q_j^{re}, \frac{A_{2j}}{A_{1j} + A_{2j}}, A_{1j} + A_{2j}) \quad \text{(Credit rating)} \\
& Q_j^{re} \leq \alpha \sum_m q(\mathbf{p}_m, \mathbf{R}) \quad \text{(Cap of reinsurance)}
\end{aligned}$$

Table 16 presents the results of the robustness check. After removing both capital requirements, the results remain similar to those reported in Table 7, suggesting that the risk-based

capital requirement plays a more important role than the fixed minimum capital requirement. When the minimum capital requirement is also removed, insurers hold substantially fewer assets. Relative to the baseline specification, total asset holdings decrease by 79.7%, or \$9.16 million on average. Insurers also hold substantially less reinsurance.

After removing capital requirements, insolvency increases, suggesting that credit ratings alone may not be sufficient to replace capital regulation. Table 13 also suggests that higher risk-based capital ratio negatively correlates with insolvency events. Moreover, the insolvency costs estimated under both definitions are likely to be underestimated, as they do not capture the broader social costs of insurer failure. These include administrative costs, disruptions to consumers from delayed or unpaid claims, difficulties obtaining replacement coverage, losses to creditors and guaranty funds, and potential spillovers to other insurers. Since property insurers are also important participants in bond markets, large-scale failures may increase asset sales and financial-market volatility. Therefore, although consumer and producer surplus appear to increase, these measured gains may be offset by unaccounted costs. Thus, we cannot conclude that eliminating capital requirements and relying solely on market discipline would improve social welfare.

Table 7: Removing capital requirement increases insolvency and affordability

Variable	Mean			Median		
	Baseline (1)	Counterfactual (2)	Diff (3)	Baseline (4)	Counterfactual (5)	Diff (6)
insolvency	0.0067	0.0074	0.0007	0.0000	0.0000	0.0000
p in risky market	1.9082	1.8849	-0.0233	1.8423	1.8165	-0.0258
p in less risky market	2.0962	1.8532	-0.2430	1.8909	1.8428	-0.0481
q in risky market	27.7211	30.0606	2.3395	6.7679	7.0184	0.2504
q in less risky market	9.2884	11.1843	1.8959	3.0043	3.2623	0.2580
profits	49.0937	56.0361	6.9425	18.8682	18.9041	0.0359
operation profits	44.5993	51.6954	7.0961	18.1630	18.2662	0.1032
investment profits	2.1994	1.8016	-0.3978	0.6268	0.6254	-0.0014
risky assets	5.2746	0.4936	-4.7809	0.0297	0.0015	-0.0281
total assets	11.4897	2.3346	-9.1551	2.0013	2.0005	-0.0008
Q_{re}	5.8704	3.8248	-2.0456	0.0023	0.0008	-0.0014
HHI in risky market	0.0115	0.0113	-0.0002	0.0105	0.0104	-0.0001
HHI in less risky market	0.0101	0.0096	-0.0005	0.0097	0.0099	0.0002
consumer surplus	585848.3701	592447.8669	6599.4968	608838.7637	614215.8847	5377.1210
producer surplus	37596.6521	43396.8815	5800.2294	36699.9523	43595.0409	6895.0886
insolvency cost	128.2633	144.0639	15.8006	125.4750	125.4750	0.0000
welfare	623316.7589	635700.6845	12383.9256	648808.4123	657695.2540	8886.8417

Notes: This table shows comparison of statistics of original data and counterfactual. The first three columns display mean of variables. The last three columns display median of variables. Columns (1) and (4) show statistics of original data. Columns (2) and (5) show results of counterfactual analysis when removing risk-based capital requirement. Column (3) is equal to column (2) minus column (1). Similarly, column (6) is equal to column (5) minus column (4). The unit of q and Q_{re} is a million. Except for insolvency, p, q, Q_{re} , and HHI, the unit of variables is a million dollars.

6.3 A World without Capital Regulation and High Rating Salience

A central question in financial regulation is whether market discipline, reinforced by financially literate consumers, can serve as a viable substitute for formal government oversight. If the salience of risk indicators like credit ratings were sufficiently high, consumers could theoretically discipline insurers into holding adequate capital, thereby mitigating insolvency risk without the need for binding capital requirements. This would be a first-best outcome, as it would reduce the potential deadweight losses associated with regulatory capital. I test this hypothesis directly within my structural framework.

In the baseline model, the estimated demand coefficient on price is -1.0157, while the coefficient on credit ratings is 0.1173. This indicates that in the current market, consumers are more responsive to price than to an insurer's credit rating. To simulate a world with powerful market discipline, I conduct a counterfactual analysis where I increase the salience of credit ratings. Specifically, I increase the rating coefficient by a factor of 1.5 to 0.176.

Table 8 presents the results for a market without capital regulation and with high credit rating salience. In the absence of a risk-based capital requirement, greater consumer sensitivity to credit ratings still reduces the insolvency rate by 0.1 percentage points relative to the baseline. The price decreases by 0.78% in the risky market, while insurance demand increases in both markets. Consequently, operating profits increase by 42%, or \$18.8 million, on average. The increase in the insolvency rate relative to the counterfactual scenario without a capital requirement and with the original level of rating salience arises because firms expand more aggressively into less risky markets.

Total asset holdings decrease relative to the baseline, although the reduction is smaller than that observed in the counterfactual scenario without a capital requirement and with the original level of rating salience. Under high credit rating salience, insurers allocate a larger share of their assets to safe assets in order to improve their credit ratings and attract consumers. They also increase their reinsurance holdings, which further improves their credit

ratings. Because greater reliance on reinsurance raises the threshold for insurers to remain active in the market, market concentration increases, as reflected by the higher Herfindahl–Hirschman Index (HHI), indicating weaker market competition.

Although insolvency costs increase by 16.67%, or \$21.38 million, aggregate welfare nevertheless increases. The gains in producer surplus and consumer surplus more than offset the increase in insolvency costs, with the resulting expansion in market scale dominating the additional costs associated with insolvency.

I conduct an additional robustness check by considering a setting with high credit-rating salience while removing both the risk-based capital requirement and the minimum capital regulation. The results are reported in Table 17. In this specification, the price in the risky market decreases further, by 0.82% relative to the baseline scenario. Overall, the results without the minimum capital requirement in Table 17 are very similar to those reported in Table 8, discussed in the preceding paragraphs.

These results suggest that high credit-rating salience cannot substitute for capital requirements. Even when credit ratings are highly salient to consumers, the insolvency rate remains higher in the absence of capital requirements. Moreover, market concentration increases under this regime. Thus, even when consumers are attentive to and responsive to credit ratings, market discipline alone is insufficient to replace capital regulation. These findings underscore the importance of maintaining capital requirements as a complement to market-based mechanisms for promoting financial stability.

Table 8: High Rating Salience Improves Affordability and Increases Market Concentration

Variable	Mean			Median		
	Baseline	Counterfactual	Diff	Baseline	Counterfactual	Diff
	(1)	(2)	(3)	(4)	(5)	(6)
insolvency	0.0067	0.0077	0.0010	0.0000	0.0000	0.0000
p in risky market	1.9082	1.8932	-0.0150	1.8423	1.8324	-0.0099
p in less risky market	2.0962	1.8550	-0.2412	1.8909	1.8481	-0.0428
q in risky market	27.7211	40.0812	12.3601	6.7679	6.7142	-0.0537
q in less risky market	9.2884	14.7235	5.4351	3.0043	3.2039	0.1996
profits	49.0937	68.8192	19.7256	18.8682	18.4314	-0.4368
operation profits	44.5993	63.3985	18.7992	18.1630	17.8657	-0.2973
investment profits	2.1994	2.3339	0.1345	0.6268	0.5656	-0.0612
risky assets	5.2746	0.3009	-4.9736	0.0297	0.0004	-0.0292
total assets	11.4897	2.6535	-8.8363	2.0013	2.0005	-0.0008
Q_{re}	5.8704	7.6040	1.7336	0.0023	0.0006	-0.0017
HHI in risky market	0.0115	0.0140	0.0025	0.0105	0.0129	0.0024
HHI in less risky market	0.0101	0.0125	0.0024	0.0097	0.0125	0.0027
consumer surplus	585848.3701	651100.9381	65252.5680	608838.7637	669559.4814	60720.7177
producer surplus	37596.6521	53136.0793	15539.4272	36699.9523	48616.6716	11916.7192
insolvency cost	128.2633	149.6406	21.3772	125.4750	150.5700	25.0950
welfare	623316.7589	704087.3769	80770.6180	648808.4123	716576.1326	67767.7203

Notes: This table shows comparison of statistics of original data and counterfactual. The first three columns display mean of variables. The last three columns display median of variables. Columns (1) and (4) show statistics of original data. Columns (2) and (5) show results of counterfactual analysis when removing risk-based capital requirement and credit rating salience is high. Columns (3) is equal to column (2) minus column (1). Similarly, column (6) is equal to column (5) minus column (4). The unit of q and Q_{re} is a million. Except for insolvency, p, q, Q_{re} , and HHI, the unit of variables is a million dollars.

7 Conclusion

In recent years, policymakers have debated whether capital regulation for U.S. property and casualty insurers should transition from a static formula-based framework to a model-based approach. Before making such a shift, it is crucial to deepen our understanding of how capital regulation affects market outcomes and what specific roles it plays in the presence of credit ratings. This paper addresses two central questions: (1) how capital regulation influences insurer solvency and market outcomes, and (2) how capital regulation constrains insolvency when credit ratings are an important market discipline mechanism.

I first provide empirical evidence on the impact of capital regulation on insurer solvency and product affordability. The analysis indicates that while capital regulation strengthens solvency, it also increases insurance prices. Specifically, a \$1 million increase in required capital leads to a \$3.098 million increase in actual capital held and a 0.095 percentage point rise in insurance premiums.

To further investigate these dynamics, I develop a theoretical framework that jointly incorporates capital requirements and credit ratings. Using structural estimation, I conduct counterfactual simulations to inform future policy design. The first counterfactual analysis produces results consistent with the reduced-form evidence: stricter capital requirements improve solvency but also raise prices. The second counterfactual analysis examines the relative roles of capital regulation and credit ratings in mitigating insolvency risk. A key question is whether credit ratings alone can sufficiently discipline insurers. The counterfactual results indicate that, in the absence of risk-based capital regulation, the insolvency rate would increase by 0.07 percentage points, while insurance premiums would decrease by about 1.2%. The average asset holdings decline to \$2.33 million, only slightly above the minimum capital requirement. The third counterfactual analysis explores a market without capital regulation but with high credit rating salience. When consumers place greater weight on credit ratings, the insolvency rate still increases. Although the prices decrease and

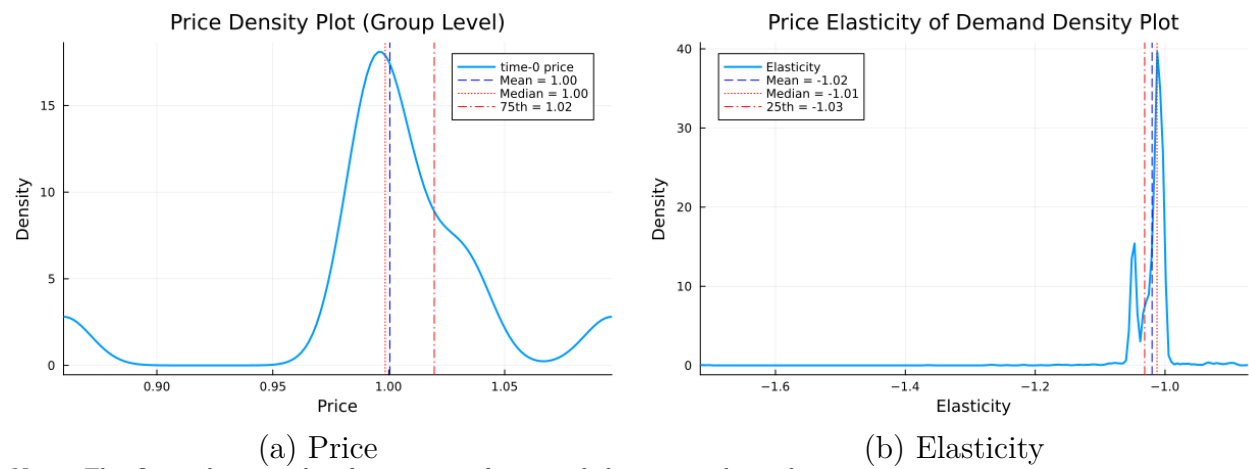
reinsurance holdings increase, market concentration increases. This finding suggests that enhancing consumer attention to credit ratings alone can not offer a perfect substitute for capital regulation. In further robustness checks, I remove minimum capital requirement as well as risk-based capital requirement, and results are similar.

Overall, the findings underscore the critical role of capital regulation in maintaining market stability. Even with robust credit rating mechanisms, capital requirements remain essential to curb excessive risk-taking and preserve solvency in the insurance sector.

8 Appendix

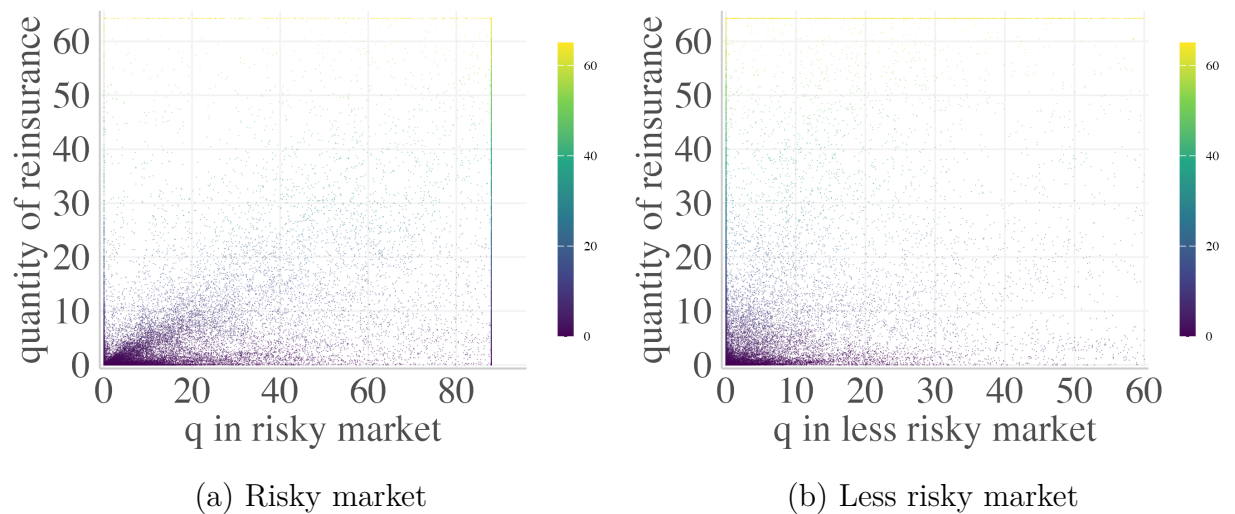
8.1 Figures

Figure 4: Estimation Results of Price and Elasticity in Demand Estimation



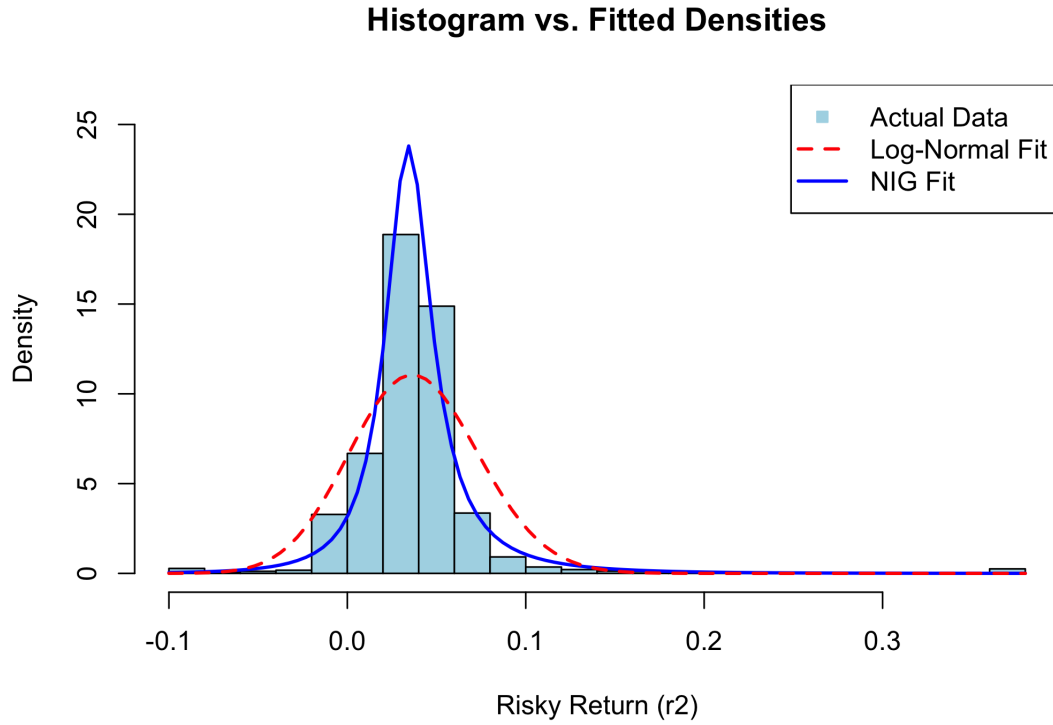
Notes: This figure shows results of estimation of price and elasticity in demand estimation.

Figure 5: Scatter Plots of Quantity of Insurance and Quantity of Reinsurance



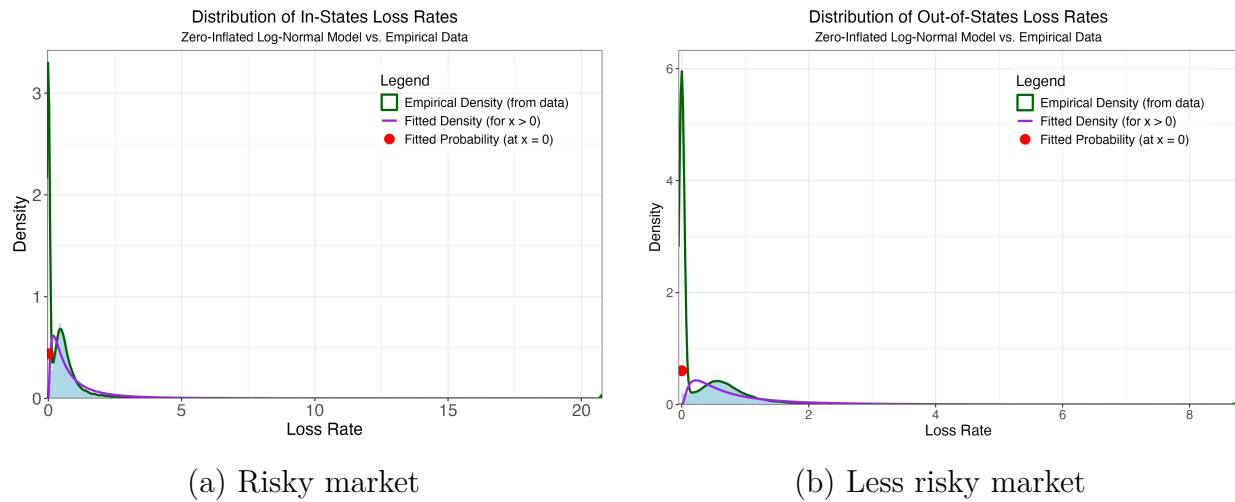
Notes: This figure shows scattered plots of quantity of insurance and quantity of reinsurance. Panel (a) shows quantity in risky markets. Panel (b) shows quantity in less risky markets.

Figure 6: Estimation Results of Return of Risky Assets



Notes: This figure shows results of estimation of return of risky assets. The normal inverse Gaussian (NIG) distribution fits the data better.

Figure 7: Estimation Results of Loss Rates



Notes: This figure shows results of estimation of loss rates. Panel (a) is the loss rate in risky market, which has exposure to hurricane and earthquake. Panel (b) is the loss rate in less risky market.

8.2 Tables

Table 9: Summary of catastrophe-prone areas

Disaster	Catastrophe-prone areas
Hurricane	Hawaii, District of Columbia and states and commonwealths bordering on the Atlantic Ocean and/or the Gulf of Mexico including Puerto Rico.
Earthquake	Alaska, Hawaii, Washington, Oregon, California, Idaho, Nevada, Utah, Arizona, Montana, Wyoming, Colorado, New Mexico, Puerto Rico, and geographic areas in the following states that are in the New Madrid Seismic Zone - Missouri, Arkansas, Mississippi, Tennessee, Illinois and Kentucky.

Notes: This table summarizes the catastrophe-prone areas which I use to define the treated regions.

Table 10: Summary statistics

	Entire sample			Treated		Control	
	mean	sd	median	before	after	before	after
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
total adjusted capital	203.4217	390.1192	50.0268	217.7305	252.9410	110.4573	122.6043
direct written premiums	251.0444	385.7734	72.0020	266.0319	328.2779	98.0190	135.2628
price change	1.1474	2.4061	0.0000	0.9909	1.4741	0.5868	0.8305
net written premiums	146.0623	309.4856	13.4776	152.9786	182.9649	73.8050	90.6193
RBC ratio	72.8141	136.5170	11.3923	74.8804	69.9115	76.4385	75.8249
reserve	132.4696	275.4144	9.0012	144.6681	167.6183	66.9479	72.9762
required capital	28.4523	62.5293	3.2278	29.7217	37.0360	12.9843	15.3261
surplus	203.7210	388.6703	50.1500	217.5556	252.4943	113.5543	123.7909
treatment	0.6857	0.4643	1.0000	1.0000	1.0000	0.0000	0.0000

Notes: This table shows summary statistics. Columns (1) to (3) show the entire sample. Columns (4) to (7) displays means. Columns (4) and (5) are treated group before year 2017 and after year 2017 respectively. Columns (6) and (7) are control group before year 2017 and after year 2017 respectively. Except for price change, RBC ratio, and treatment, the unit of variables is million. The unit of price change and RBC ratio is percent.

Table 11: First stage (binding)

	<i>Dependent variable:</i>					
	required capital	surplus	price change	required capital	surplus	price change
	(1)	(2)	(3)	(4)	(5)	(6)
treated×reform	3.536*** (1.007)	11.199*** (4.317)	0.409*** (0.117)	3.351*** (0.730)	11.420*** (3.282)	0.336*** (0.089)
Observations	5,920	5,920	5,920	9,665	9,665	9,665
R ²	0.0378	0.0387	0.0636	0.0397	0.0427	0.0673

Note:

*p<0.1; **p<0.05; ***p<0.01

Notes: This table displays reduced-form results of binding sample which is defined as average RBC ratio before treatment is within 1 or 2 standard deviations to threshold 2. Columns (1), (2), and (3): Average RBC ratio before treatment is within 1 standard deviation to threshold 2. Columns (4), (5) and (6): Average RBC ratio before treatment is within 2 standard deviations to threshold 2.

Table 12: IV regressions (binding)

	<i>Dependent variable:</i>			
	surplus	price change	surplus	price change
	(1)	(2)	(3)	(4)
required capital	3.167*** (0.962)	0.116** (0.046)	3.408*** (0.768)	0.100*** (0.034)
Observations	5,920	5,920	9,665	9,665
R ²	0.392	0.0636	0.401	0.0673

Note:

*p<0.1; **p<0.05; ***p<0.01

Notes: This table displays IV regression results of binding sample which is defined as average RBC ratio before treatment is within 1 or 2 standard deviations to threshold 2. Columns (1) and (2): Average RBC ratio before treatment is within 1 standard deviation to threshold 2. Columns (3) and (4): Average RBC ratio before treatment is within 2 standard deviations to threshold 2.

Table 13: Higher RBC ratios lead to lower probability of insolvency

	<i>Dependent variable:</i>			
	insolvency			
	<i>OLS</i>		<i>panel linear</i>	
	(1)	(2)	(3)	(4)
RBC ratio	−0.0001*** (0.00001)	−0.0001*** (0.00001)	−0.00003*** (0.00000)	−0.00002*** (0.00000)
Constant	0.054*** (0.001)			
Observations	61,223	61,223	61,223	61,223
R ²	0.002	0.002	0.001	0.0003

Note:

*p<0.1; **p<0.05; ***p<0.01

Notes: This table shows that higher RBC ratios lead to lower probability of insolvency. Columns (2) and (3): Control for time fixed effects. Columns (3) and (4): Set insolvency to be 1 when it is 3 years before being insolvent. Column (4): Control for time and firm fixed effects.

Table 14: Estimation Results of Capital Regulation Function

Variable	Dependent Variable: $\tilde{K}(\cdot)$	
	Before reform	After reform
q in risky market	0.0245*** (0.0036)	0.0312*** (0.0071)
q in less risky market	0.0000 (0.0008)	0.0000 (0.0000)
Risky Asset	0.0777*** (0.0003)	0.0805*** (0.0005)
Q_{re}	-0.0000 (0.0000)	-0.0000 (0.0072)
Observations	31,706	9,885
In-Sample R^2	0.8813	0.8823
Out-of-Sample R^2	0.8811	0.8757

. $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: I use linear regression to estimate capital regulation function. Column (1) shows estimation for capital regulation function before the 2017 reform, and Column (2) represents results after the reform.

Table 15: Estimation Results of Credit Rating Function

Variable	Dependent Variable: R_j (Credit Rating)	
	Before reform	After reform
Intercept	16.8384*** (0.1051)	17.9971*** (0.0833)
q in risky market	0.0000 (0.0000)	0.0000 (0.0000)
q in less risky market	0.0077** (0.0023)	0.0052*** (0.0011)
Total Asset	0.0002*** (0.0000)	0.0000*** (0.0000)
Risky Asset Share	-0.1671 (0.1267)	-1.0434*** (0.1047)
Q_{re}	0.0514*** (0.0030)	0.0338*** (0.0012)
Observations	13,631	24,753
Out-of-Sample R^2	0.0697	0.0496
OOS ± 1 Notch Accuracy	48.84%	48.09%

. $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Notes: I use linear regression to estimate credit rating function. Column (1) shows estimation for credit rating function before 2005 reform, and Column (2) represents results after the reform.

Table 16: Removing capital requirement increases insolvency and affordability (risk-based and minimum capital requirements)

Variable	Mean			Median		
	Baseline	Counterfactual	Diff	Baseline	Counterfactual	Diff
	(1)	(2)	(3)	(4)	(5)	(6)
insolvency	0.0067	0.0074	0.0007	0.0000	0.0000	0.0000
p in risky market	1.9082	1.8878	-0.0205	1.8423	1.8201	-0.0222
p in less risky market	2.0962	1.8559	-0.2403	1.8909	1.8439	-0.0470
q in risky market	27.7211	29.9904	2.2693	6.7679	6.9193	0.1514
q in less risky market	9.2884	11.1417	1.8533	3.0043	3.2529	0.2485
profits	49.0937	55.9021	6.8084	18.8682	18.8440	-0.0242
operation profits	44.5993	51.5708	6.9715	18.1630	18.2401	0.0771
investment profits	2.1994	1.7971	-0.4023	0.6268	0.6246	-0.0022
risky assets	5.2746	0.4920	-4.7826	0.0297	0.0013	-0.0283
total assets	11.4897	2.3324	-9.1573	2.0013	2.0005	-0.0008
Q_{re}	5.8704	3.8054	-2.0649	0.0023	0.0008	-0.0015
HHI in risky market	0.0115	0.0113	-0.0002	0.0105	0.0103	-0.0002
HHI in less risky market	0.0101	0.0096	-0.0005	0.0097	0.0099	0.0002
consumer surplus	585848.3701	592448.2342	6599.8641	608838.7637	614136.5064	5297.7427
producer surplus	37596.6521	43373.8130	5777.1609	36699.9523	43553.5715	6853.6192
insolvency cost	128.2633	144.0639	15.8006	125.4750	125.4750	0.0000
welfare	623316.7589	635677.9833	12361.2244	648808.4123	657564.6030	8756.1907

Notes: This table shows comparison of statistics of original data and counterfactual. The first three columns display mean of variables. The last three columns display median of variables. Columns (1) and (4) show statistics of original data. Columns (2) and (5) show results of counterfactual analysis when removing risk-based and minimum capital requirements. Column (3) is equal to column (2) minus column (1). Similarly, column (6) is equal to column (5) minus column (4). The unit of q and Q_{re} is a million. Except for insolvency, p, q, Q_{re} , and HHI, the unit of variables is a million dollars.

Table 17: High Rating Salience Improves Affordability and Increases Market Concentration (risk-based and minimum capital requirements)

Variable	Mean			Median		
	Baseline (1)	Counterfactual (2)	Diff (3)	Baseline (4)	Counterfactual (5)	Diff (6)
insolvency	0.0067	0.0078	0.0011	0.0000	0.0000	0.0000
p in risky market	1.9082	1.8925	-0.0157	1.8423	1.8294	-0.0129
p in less risky market	2.0962	1.8529	-0.2433	1.8909	1.8450	-0.0459
q in risky market	27.7211	39.9714	12.2503	6.7679	6.6760	-0.0919
q in less risky market	9.2884	14.6810	5.3926	3.0043	3.2081	0.2037
profits	49.0937	68.6494	19.5557	18.8682	18.4342	-0.4340
operation profits	44.5993	63.2408	18.6414	18.1630	17.8487	-0.3143
investment profits	2.1994	2.3291	0.1296	0.6268	0.5660	-0.0608
risky assets	5.2746	0.2983	-4.9763	0.0297	0.0006	-0.0291
total assets	11.4897	2.6535	-8.8362	2.0013	2.0006	-0.0007
Q_{re}	5.8704	7.5682	1.6978	0.0023	0.0007	-0.0015
HHI in risky market	0.0115	0.0140	0.0025	0.0105	0.0129	0.0024
HHI in less risky market	0.0101	0.0125	0.0024	0.0097	0.0125	0.0028
consumer surplus	585848.3701	651093.0389	65244.6688	608838.7637	669471.0513	60632.2876
producer surplus	37596.6521	53106.6450	15509.9929	36699.9523	48445.2721	11745.3198
insolvency cost	128.2633	150.5700	22.3067	125.4750	150.5700	25.0950
welfare	623316.7589	704049.1139	80732.3550	648808.4123	716410.3428	67601.9305

Notes: This table shows comparison of statistics of original data and counterfactual. The first three columns display mean of variables. The last three columns display median of variables. Columns (1) and (4) show statistics of original data. Columns (2) and (5) show results of counterfactual analysis when removing risk-based and minimum capital requirements, and when credit rating salience is high. Columns (3) is equal to column (2) minus column (1). Similarly, column (6) is equal to column (5) minus column (4). The unit of q and Q_{re} is a million. Except for insolvency, p, q, Q_{re} , and HHI, the unit of variables is a million dollars.

Table 18: Comparison of insolvency costs and welfare using different definitions

Scenario	Definition 1		Definition 2	
	Insolvency cost	Welfare	Insolvency cost	Welfare
Baseline	128.2633	623316.8	1810.095	621634.9
High RBC threshold	100.3800	617370.3	1381.994	616088.7
No Risk-based cap	144.0639	635700.7	2026.069	633818.7
No Risk-based cap & High credit salience	149.6406	704087.4	2445.088	701791.9
No Risk-based cap & No minimum cap	144.0639	635678.0	2025.898	633796.1
No Risk-based cap, No minimum cap & High credit salience	150.5700	704049.1	2444.044	701755.6

Notes: This table shows comparison of insolvency costs and welfare using different definitions of insolvency costs. The first two columns display results using the default definition of insolvency costs, which is number of insolvency times average payment for per insolvency from Guaranty Funds. The last two columns display results using the second definition, which is realized loss minus total assets. The unit of variables is a million dollars.

8.3 Development of Capital Requirement before 2017

According to Klein (1995), several property-casualty insurers in the United States failed during the 1980s due to deficient loss reserves, inadequate pricing, and overly rapid expansion. The fixed capital requirements in place at the time were insufficient to constrain this rapid growth, which contributed significantly to these failures. In response, the National Association of Insurance Commissioners (NAIC) developed and implemented Risk-Based Capital (RBC) formulas in the early 1990s, allowing capital requirements to vary with firm-specific risk profiles and size. The RBC formula for property and casualty insurers was introduced in 1994. The RBC standards for life and property/casualty (P/C) insurers are based on the Risk-Based Capital (RBC) for Insurers Model Act (Model #312), originally adopted by the NAIC in 1993 and most recently revised in 2011.

Beginning January 1, 2015, the NAIC implemented the Own Risk and Solvency Assessment (ORSA) Model Act, which mandates that insurers conduct and annually report on their enterprise risk management (ERM) practices and capital adequacy. ERM refers to a comprehensive, organization-wide framework for identifying and managing risk exposures.

8.4 Derivation of First-Order Conditions for Supply Estimation

In the structural estimation of ν_j^d , I assume that the cap of reinsurance holding constraint is not binding. Since $\iiint_{\Omega} = \iiint_{\Omega/D} + \iiint_D$, then objective function becomes:

$$\begin{aligned} & \max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \iiint_{D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} \left[\sum_m ((p_{jm} - L_{jm} - Exp_j)q(\mathbf{p}_m, \mathbf{R})) - p^{re} Q_j^{re} \right. \\ & - FC_j + \min\{CQ_j^{re}, \max\{0, \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d_j\}\} + r_1(A_{1j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R}) \frac{A_{1j}}{A_{1j} + A_{2j}}) + \\ & r_2(A_{2j} + \sum_m p_{jm}q(\mathbf{p}_m, \mathbf{R}) \frac{A_{2j}}{A_{1j} + A_{2j}})] f dV \\ & - \iiint_{\Omega/D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} (A_{1j} + A_{2j} + \nu_j^d) f dV - r_c(A_{1j} + A_{2j}) \end{aligned}$$

Plug in the expression of the payment from reinsurance.

$$\begin{aligned}
& \max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \iiint_{D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} [\sum_m (p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R}) - p^{re} Q_j^{re} - FC_j \\
& - \frac{1}{k} \log \left(\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))} \right) \\
& + r_1(A_{1j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{1j}}{A_{1j} + A_{2j}}) + r_2(A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{2j}}{A_{1j} + A_{2j}})] f_r(r_2) f_{L_1}(L_{j1}) f_{L_2}(L_{j2}) dr_2 dL_{j1} dL_{j2} \\
& - \iiint_{\Omega} (A_{1j} + A_{2j} + \nu_j^d) f_r(r_2) f_{L_1}(L_{j1}) f_{L_2}(L_{j2}) dr_2 dL_{j1} dL_{j2} \\
& + \iiint_{D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} (A_{1j} + A_{2j} + \nu_j^d) f_r(r_2) f_{L_1}(L_{j1}) f_{L_2}(L_{j2}) dr_2 dL_{j1} dL_{j2} - r_c(A_{1j} + A_{2j})
\end{aligned}$$

Further simplify objective function:

$$\begin{aligned}
& \max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \iiint_{D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} [\sum_m ((p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R})) - p^{re} Q_j^{re} - FC_j \\
& - \frac{1}{k} \log \left(\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))} \right) \\
& + r_1(A_{1j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{1j}}{A_{1j} + A_{2j}}) + r_2(A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{2j}}{A_{1j} + A_{2j}}) \\
& + A_{1j} + A_{2j} + \nu_j^d] f_r(r_2) f_{L_1}(L_{j1}) f_{L_2}(L_{j2}) dr_2 dL_{j1} dL_{j2} - \nu_j^d - (1 + r_c)(A_{1j} + A_{2j})
\end{aligned}$$

Let's call $dr_2 dL_{j1} dL_{j2}$ as dV , and $f_r(r_2) f_{L_1}(L_{j1}) f_{L_2}(L_{j2})$ as f . Then,

$$\begin{aligned}
& \max_{\{p_{jm}, A_{1j}, A_{2j}, Q_j^{re}\}} \iiint_{D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})} [\sum_m ((p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R})) - p^{re} Q_j^{re} - FC_j \\
& - \frac{1}{k} \log \left(\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))} \right) \\
& + r_1(A_{1j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{1j}}{A_{1j} + A_{2j}}) + r_2(A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{2j}}{A_{1j} + A_{2j}}) \\
& + A_{1j} + A_{2j} + \nu_j^d] f dV - \nu_j^d - (1 + r_c)(A_{1j} + A_{2j})
\end{aligned}$$

s.t.

$$R_j = g(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), \frac{A_{2j}}{A_{1j} + A_{2j}}, A_{1j} + A_{2j}, Q_j^{re}) \quad (\text{Credit rating})$$

$$2 \leq \frac{A_{1j} + A_{2j}}{\bar{K}(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), A_{2j}, Q_j^{re})} \quad (\text{Risk-based capital requirement})$$

$$C_{\min} \leq A_{1j} + A_{2j} \quad (\text{Minimum capital requirement})$$

The domain $D(p_{jm}, A_{1j}, A_{2j}, Q_j^{re})$ where insurers are solvent is

$$\begin{aligned} & [\sum_m (p_{jm} - L_{jm} - Exp_j) q(\mathbf{p}_m, \mathbf{R}) - p^{re} Q_j^{re} - FC_j + r_1(A_{1j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{1j}}{A_{1j} + A_{2j}}) + \\ & r_2(A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \frac{A_{2j}}{A_{1j} + A_{2j}}) - \frac{1}{k} \log \left(\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))} \right)] \\ & \geq -(A_{1j} + A_{2j} + \nu_j^d) \end{aligned}$$

$M = 2$ in this case.

In this section, I am going break down derivatives by parts. First, I take derivative of the demand function to p_{jm} , we have:

$$\frac{\partial q_{jm}}{\partial p_{jm}} = N_m \alpha_1 S_{jm} (1 - S_{jm})$$

Similarly, take derivative of the demand function to R , we have:

$$\frac{\partial q_{jm}}{\partial R_j} = N_m \alpha_2 S_{jm} (1 - S_{jm})$$

$$N_m = \sum_k q_{km}$$

N_m is market size in market m . S_{jm} is market share.

Let the total premium collected be $P_{tot} = \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R})$ and the blended float yield be $\Phi_j = \frac{r_1 A_{1j} + r_2 A_{2j}}{A_{1j} + A_{2j}}$. Also set

$$\Psi_j = \frac{\frac{\exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))}{(1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d)))^2}}{\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))}}$$

Set the RBC multiplier as λ , and the minimum capital multiplier as μ .

FOC w.r.t p_{j1} :

$$\iiint_D \left[\frac{\partial G}{\partial p_{j1}} + \frac{\partial G}{\partial R_j} \frac{\partial R_j}{\partial p_{j1}} \right] f dV - 2\lambda \left[\frac{\partial \tilde{K}}{\partial q_1} \frac{\partial q_1}{\partial p_{j1}} + \frac{\partial \tilde{K}}{\partial R_j} \frac{\partial R_j}{\partial p_{j1}} \right] = 0$$

These are expressions of each components.

$$\begin{aligned} \frac{\partial G}{\partial p_{j1}} &= q(\mathbf{p}_1, \mathbf{R}) + \frac{\partial q(\mathbf{p}_1, \mathbf{R})}{\partial p_{j1}} (p_{j1} - L_{j1} - \text{Exp}_j) + \Psi_j L_{j1} \frac{\partial q(\mathbf{p}_1, \mathbf{R})}{\partial p_{j1}} + \left(q(\mathbf{p}_1, \mathbf{R}) + p_{j1} \frac{\partial q(\mathbf{p}_1, \mathbf{R})}{\partial p_{j1}} \right) \Phi_j \\ \frac{\partial G}{\partial R_j} &= \sum_m \frac{\partial q(\mathbf{p}_m, \mathbf{R})}{\partial R_j} (p_{jm} - L_{jm} - \text{Exp}_j) + \Psi_j \sum_m \left(L_{jm} \frac{\partial q(\mathbf{p}_m, \mathbf{R})}{\partial R_j} \right) + \left(\sum_m p_{jm} \frac{\partial q(\mathbf{p}_m, \mathbf{R})}{\partial R_j} \right) \Phi_j \\ \frac{\partial \tilde{K}}{\partial R_j} &= \frac{\partial \tilde{K}}{\partial (q(\mathbf{p}_1, \mathbf{R}))} \frac{\partial q(\mathbf{p}_1, \mathbf{R})}{\partial R_j} + \frac{\partial \tilde{K}}{\partial (q(\mathbf{p}_2, \mathbf{R}))} \frac{\partial q(\mathbf{p}_2, \mathbf{R})}{\partial R_j} \end{aligned}$$

The FOC for each decision variable x is given by $\frac{\partial \mathcal{L}}{\partial x} = 0$, which requires applying the chain rule through the implicitly defined function R_j . The derivative of R_j is found using the Implicit Function Theorem: $\frac{\partial R_j}{\partial x} = \frac{\partial g / \partial x}{1 - \partial g / \partial R_j}$.

$$\begin{aligned} \frac{\partial g}{\partial p_{j1}} &= g_1 \left(\frac{\partial q_1}{\partial p_{j1}} \right) \\ \frac{\partial g}{\partial R_j} &= g_1 \frac{\partial q_1}{\partial R_j} + g_2 \frac{\partial q_2}{\partial R_j} \end{aligned}$$

FOC w.r.t p_{j2} :

$$\iiint_D \left[\frac{\partial G}{\partial p_{j2}} + \frac{\partial G}{\partial R_j} \frac{\partial R_j}{\partial p_{j2}} \right] f dV - 2\lambda \left[\frac{\partial \tilde{K}}{\partial q_2} \frac{\partial q_2}{\partial p_{j2}} + \frac{\partial \tilde{K}}{\partial R_j} \frac{\partial R_j}{\partial p_{j2}} \right] = 0$$

$$\frac{\partial G}{\partial p_{j2}} = q(\mathbf{p}_2, \mathbf{R}) + \frac{\partial q(\mathbf{p}_2, \mathbf{R})}{\partial p_{j2}} (p_{j2} - L_{j2} - Exp_j) + \Psi_j L_{j2} \frac{\partial q(\mathbf{p}_2, \mathbf{R})}{\partial p_{j2}} + \left(q(\mathbf{p}_2, \mathbf{R}) + p_{j2} \frac{\partial q(\mathbf{p}_2, \mathbf{R})}{\partial p_{j2}} \right) \Phi_j$$

$$\frac{\partial g}{\partial p_{j2}} = g_2 \left(\frac{\partial q_2}{\partial p_{j2}} \right)$$

FOC for A_{1j} :

$$\iiint_D \left[(r_1 + 1) + P_{tot} \frac{A_{2j}(r_1 - r_2)}{(A_{1j} + A_{2j})^2} + \frac{\partial G}{\partial R_j} \frac{\partial R_j}{\partial A_{1j}} \right] f dV - (1 + r_c) - \lambda \left[2 \left(\frac{\partial \tilde{K}}{\partial R_j} \frac{\partial R_j}{\partial A_{1j}} \right) - 1 \right] + \mu = 0$$

$$\frac{\partial g}{\partial A_{1j}} = - \frac{\partial g}{\partial \left(\frac{A_{2j}}{A_{1j} + A_{2j}} \right)} \left(\frac{A_{2j}}{(A_{1j} + A_{2j})^2} \right) + \frac{\partial g}{\partial (A_{1j} + A_{2j})}$$

FOC for A_{2j} :

$$\iiint_D \left[(r_2 + 1) + P_{tot} \frac{A_{1j}(r_2 - r_1)}{(A_{1j} + A_{2j})^2} + \frac{\partial G}{\partial R_j} \frac{\partial R_j}{\partial A_{2j}} \right] f dV - (1 + r_c) - \lambda \left[2 \left(\frac{\partial \tilde{K}}{\partial A_{2j}} + \frac{\partial \tilde{K}}{\partial R_j} \frac{\partial R_j}{\partial A_{2j}} \right) - 1 \right] + \mu = 0$$

$$\frac{\partial g}{\partial A_{2j}} = \frac{\partial g}{\partial \left(\frac{A_{2j}}{A_{1j} + A_{2j}} \right)} \left(\frac{A_{1j}}{(A_{1j} + A_{2j})^2} \right) + \frac{\partial g}{\partial (A_{1j} + A_{2j})}$$

FOC for Q_j^{re} :

$$\iiint_D \left[\frac{\partial G}{\partial Q_j^{re}} + \frac{\partial G}{\partial R_j} \frac{\partial R_j}{\partial Q_j^{re}} \right] f dV - 2\lambda \left[\frac{\partial \tilde{K}}{\partial Q_j^{re}} + \frac{\partial \tilde{K}}{\partial R_j} \frac{\partial R_j}{\partial Q_j^{re}} \right] = 0$$

$$\frac{\partial G}{\partial Q_j^{re}} = -p^{re} + \frac{C \exp(-k \cdot C Q_j^{re})}{\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))}}$$

$$\frac{\partial g}{\partial Q_j^{re}} = \frac{\partial g}{\partial Q_j^{re}}$$

Primal Feasibility:

$$2 - \frac{A_{1j} + A_{2j}}{\tilde{K}(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), A_{2j})} \leq 0$$

$$A_{1j} + A_{2j} - C_{\min} \geq 0$$

Dual Feasibility:

$$\lambda \geq 0$$

$$\mu \geq 0$$

Complementary Slackness:

$$\lambda \left(2 - \frac{A_{1j} + A_{2j}}{\tilde{K}(q(\mathbf{p}_1, \mathbf{R}), q(\mathbf{p}_2, \mathbf{R}), A_{2j})} \right) = 0$$

$$\mu(A_{1j} + A_{2j} - C_{\min}) = 0$$

8.5 Construct Moment Condition

$$\because \text{Quantity share } S_{jmt} = \frac{q_{jmt}}{\sum_{k \in \mathcal{C}} q_{kmt}} = \frac{e^{u_{jmt}}}{\sum_{k \in \mathcal{C}} e^{u_{kmt}}}$$

$$\therefore \log S_{jmt} = u_{jmt} - \log \sum_{k \in \mathcal{C}} e^{u_{kt}}$$

$$\begin{aligned} \therefore \log S_{jmt} - \log S_{j'mt} &= u_{jmt} - u_{j'mt} \\ &= \alpha_1(p_{jmt} - p_{j'mt}) + \alpha_2(R_{jt} - R_{j't}) + (\xi_{jmt} - \xi_{j'mt}) \end{aligned}$$

$$\text{Then } \xi_{jmt} - \xi_{j'mt} = \log S_{jmt} - \log S_{j'mt} - \alpha_1(p_{jmt} - p_{j'mt}) - \alpha_2(R_{jt} - R_{j't})$$

Then if take time difference,

$$\begin{aligned} &(\xi_{jmt} - \xi_{jm,t-1}) - (\xi_{j'mt} - \xi_{j'm,t-1}) \\ &= (\log S_{jmt} - \log S_{jm,t-1}) - (\log S_{j'mt} - \log S_{j'm,t-1}) \\ &\quad - \alpha_1[(p_{jmt} - p_{jm,t-1}) - (p_{j'mt} - p_{j'm,t-1})] - \alpha_2[(R_{jt} - R_{j,t-1}) - (R_{j't} - R_{j',t-1})] \end{aligned}$$

$$\because \text{I only observe revenue share in the data } \omega_{jmt} = \frac{p_{jmt}q_{jmt}}{\sum_k p_{kmt}q_{kmt}}$$

$$\begin{aligned} &\therefore (\log \omega_{jmt} - \log \omega_{jm,t-1}) - (\log \omega_{j'mt} - \log \omega_{j'm,t-1}) \\ &= \alpha_1[(p_{jmt} - p_{jm,t-1}) - (p_{j'mt} - p_{j'm,t-1})] + \alpha_2[(R_{jt} - R_{j,t-1}) - (R_{j't} - R_{j',t-1})] \\ &\quad + (\xi_{jmt} - \xi_{jm,t-1}) - (\xi_{j'mt} - \xi_{j'm,t-1}) + (\log p_{jmt} - \log p_{jm,t-1}) \\ &\quad - (\log p_{j'mt} - \log p_{j'm,t-1}) \end{aligned}$$

$$\therefore d_{jmt} = \prod_{s=1}^t \frac{p_{jms}}{p_{jm,s-1}}$$

$$\therefore p_{jmt} = d_{jmt} p_{jm0}$$

$$\begin{aligned} & \therefore (\log \omega_{jmt} - \log \omega_{jm,t-1}) - (\log \omega_{j'mt} - \log \omega_{j'm,t-1}) \\ &= \alpha_1 [(d_{jmt} p_{jm0} - d_{jm,t-1} p_{jm0}) - (d_{j'mt} p_{j'm0} - d_{j'm,t-1} p_{j'm0})] \\ & \quad + \alpha_2 [(R_{jt} - R_{j,t-1}) - (R_{j't} - R_{j',t-1})] + (\xi_{jmt} - \xi_{jm,t-1}) - (\xi_{j'mt} - \xi_{j'm,t-1}) \\ & \quad + (\log p_{jmt} - \log p_{jm,t-1}) - (\log p_{j'mt} - \log p_{j'm,t-1}) \end{aligned}$$

Set $\tilde{\alpha}_{1j} = \alpha_1 p_{jm0}$, $\tilde{\alpha}_{1j'} = \alpha_1 p_{j'm0}$, then

$$\begin{aligned} & (\xi_{jmt} - \xi_{jm,t-1}) - (\xi_{j'mt} - \xi_{j'm,t-1}) \\ &= (\log \omega_{jmt} - \log \omega_{jm,t-1}) - (\log \omega_{j'mt} - \log \omega_{j'm,t-1}) - \tilde{\alpha}_{1jm} (d_{jmt} - d_{jm,t-1}) \\ & \quad + \tilde{\alpha}_{1j'm} (d_{j'mt} - d_{j'm,t-1}) - (\log d_{jmt} - \log d_{jm,t-1}) + (\log d_{j'mt} - \log d_{j'm,t-1}) \\ & \quad - \alpha_2 [(R_{jt} - R_{j,t-1}) - (R_{j't} - R_{j',t-1})] \end{aligned}$$

8.6 Derivation of derivative of the FOC

In this appendix, I prove that the derivative of the ex-ante first-order condition (FOC) with respect to the unobserved franchise value ν_j^d reduces to a surface integral over the default boundary ∂D_j :

$$\frac{\partial \mathcal{F}_x(\mathbf{x}_j; \nu_j^d)}{\partial \nu_j^d} = \iint_{W_j(\mathbf{V}) = -\nu_j^d} \nabla_x W_j(\mathbf{V}) \cdot \frac{f(\mathbf{V})}{\|\nabla_{\mathbf{V}} W_j(\mathbf{V})\|} dS$$

Let $\mathbf{V} = (L_{j1}, L_{j2}, r_2) \in \mathbb{R}^3$ denote the vector of state variables with joint probability density $f(\mathbf{V})$. The solvency region $D_j(\nu_j^d)$ is defined as the super-level set of the pre-default

wealth function:

$$D_j(\nu_j^d) = \left\{ \mathbf{V} \in \mathbb{R}^3 \mid g(\mathbf{V}, \nu_j^d) \geq 0 \right\}, \quad \text{where} \quad g(\mathbf{V}, \nu_j^d) \equiv W_j(\mathbf{V}) + \nu_j^d$$

The default boundary $\partial D_j(\nu_j^d)$ is the 2-dimensional manifold (surface) where terminal wealth equals the default threshold:

$$\partial D_j(\nu_j^d) = \left\{ \mathbf{V} \in \mathbb{R}^3 \mid W_j(\mathbf{V}) = -\nu_j^d \right\}$$

For any decision variable $x \in \{p_{j1}, p_{j2}, Q_j^{re}, A_{1j}, A_{2j}\}$, the ex-ante FOC is given by:

$$\mathcal{F}_x(\mathbf{x}_j; \nu_j^d) \equiv \iiint_{D_j(\nu_j^d)} h_x(\mathbf{V}) d\mathbf{V} - \text{MC}(x) = 0$$

where $h_x(\mathbf{V}) \equiv \frac{\partial W_j(\mathbf{V})}{\partial x} f(\mathbf{V})$ is the marginal payoff density, and $\text{MC}(x)$ is the marginal cost (which does not depend on ν_j^d).

By the general Multivariate Leibniz Integral Rule in $\mathbb{R}^n$, the derivative of a volume integral over a parameter-dependent domain $D(t)$ is given by the sum of an interior volume integral and a boundary surface integral:

$$\frac{d}{dt} \iiint_{D(t)} h(\mathbf{V}, t) d\mathbf{V} = \iiint_{D(t)} \frac{\partial h(\mathbf{V}, t)}{\partial t} d\mathbf{V} + \iint_{\partial D(t)} h(\mathbf{V}, t) (\mathbf{v} \cdot \mathbf{n}_{\text{out}}) dS \quad (14)$$

where:

- $\mathbf{n}_{\text{out}}$ is the outward-pointing unit normal vector along the boundary surface $\partial D(t)$.
- $\mathbf{v} \equiv \frac{d\mathbf{V}}{dt}$ represents the boundary expansion velocity vector as the parameter t changes.
- dS is the standard Riemannian surface area element.

Applying Equation 14 with parameter $t = \nu_j^d$, I first evaluate the interior volume

derivative:

$$\iiint_{D_j(\nu_j^d)} \frac{\partial h_x(\mathbf{V})}{\partial \nu_j^d} d\mathbf{V} = \iiint_{D_j(\nu_j^d)} \frac{\partial}{\partial \nu_j^d} \left[\frac{\partial W_j(\mathbf{V})}{\partial x} f(\mathbf{V}) \right] d\mathbf{V}$$

Because neither the final wealth function $W_j(\mathbf{V})$ nor the distribution density $f(\mathbf{V})$ depend on the unobserved insolvency shocks ν_j^d , the partial derivative of the integrand with respect to ν_j^d is zero:

$$\frac{\partial}{\partial \nu_j^d} \left[\frac{\partial W_j(\mathbf{V})}{\partial x} f(\mathbf{V}) \right] = 0 \implies \iiint_{D_j(\nu_j^d)} 0 d\mathbf{V} = 0$$

Hence, the derivative therefore reduces to the boundary surface integral:

$$\frac{\partial \mathcal{F}_x(\mathbf{x}_j; \nu_j^d)}{\partial \nu_j^d} = \iint_{\partial D_j(\nu_j^d)} \frac{\partial W_j(\mathbf{V})}{\partial x} f(\mathbf{V}) (\mathbf{v} \cdot \mathbf{n}_{\text{out}}) dS \quad (15)$$

To solve for the geometric term $\mathbf{v} \cdot \mathbf{n}_{\text{out}}$, I analyze the level-set condition that defines the boundary surface:

$$g(\mathbf{V}(\nu_j^d), \nu_j^d) \equiv W_j(\mathbf{V}(\nu_j^d)) + \nu_j^d = 0$$

Inside the solvency domain D_j , $g(\mathbf{V}, \nu_j^d) \geq 0$, while outside in default states, $g(\mathbf{V}, \nu_j^d) < 0$. Therefore, the gradient $\nabla_{\mathbf{V}} g = \nabla_{\mathbf{V}} W_j(\mathbf{V})$ points inward toward the interior of D_j . The outward-pointing unit normal vector $\mathbf{n}_{\text{out}}$ points in the direction of decreasing g :

$$\mathbf{n}_{\text{out}} = -\frac{\nabla_{\mathbf{V}} W_j(\mathbf{V})}{\|\nabla_{\mathbf{V}} W_j(\mathbf{V})\|} \quad (16)$$

Differentiating the implicit boundary condition $W_j(\mathbf{V}(\nu_j^d)) + \nu_j^d = 0$ with respect to ν_j^d via the multivariate chain rule:

$$\begin{aligned} \nabla_{\mathbf{V}} W_j(\mathbf{V}) \cdot \frac{d\mathbf{V}}{d\nu_j^d} + \frac{\partial g}{\partial \nu_j^d} &= 0 \\ \nabla_{\mathbf{V}} W_j(\mathbf{V}) \cdot \mathbf{v} + 1 &= 0 \implies \mathbf{v} \cdot \nabla_{\mathbf{V}} W_j(\mathbf{V}) = -1 \end{aligned} \quad (17)$$

Taking the scalar product of the boundary velocity vector $\mathbf{v}$ with the outward unit normal $\mathbf{n}_{\text{out}}$ using Equations 16 and 17:

$$\mathbf{v} \cdot \mathbf{n}_{\text{out}} = \mathbf{v} \cdot \left(-\frac{\nabla_{\mathbf{v}} W_j(\mathbf{V})}{\|\nabla_{\mathbf{v}} W_j(\mathbf{V})\|} \right) = -\frac{\mathbf{v} \cdot \nabla_{\mathbf{v}} W_j(\mathbf{V})}{\|\nabla_{\mathbf{v}} W_j(\mathbf{V})\|} = -\frac{-1}{\|\nabla_{\mathbf{v}} W_j(\mathbf{V})\|} = \frac{1}{\|\nabla_{\mathbf{v}} W_j(\mathbf{V})\|}$$

Substituting $\mathbf{v} \cdot \mathbf{n}_{\text{out}} = \frac{1}{\|\nabla_{\mathbf{v}} W_j(\mathbf{V})\|}$ into Equation 15 completes the proof:

$$\frac{\partial \mathcal{F}_x(\mathbf{x}_j; \nu_j^d)}{\partial \nu_j^d} = \iint_{W_j(\mathbf{V}) = -\nu_j^d} \nabla_x W_j(\mathbf{V}) \cdot \frac{f(\mathbf{V})}{\|\nabla_{\mathbf{v}} W_j(\mathbf{V})\|} dS$$

8.7 Proof of $\nabla_x W_j \neq 0$

To prove that the structural model uniquely point-identifies ν_j^d for risky firms, we must verify that the boundary gradient $\nabla_x W_j(\mathbf{V})$ remains strictly non-zero. Let $P_{\text{tot}} = \sum_m p_{jm} q_{jm}$ be the total premium collected and $\Phi_j = \frac{r_1 A_{1j} + r_2 A_{2j}}{A_{1j} + A_{2j}}$ be the blended float yield. I evaluate the total derivative for each decision variable $x \in \{p_{j1}, Q_j^{re}, A_{1j}, A_{2j}\}$ on the catastrophic boundary where losses are extreme ($L \gg p$) and reinsurance payout part is exhausted ($\Psi_j \rightarrow 0$ when L is large). The pre-default terminal wealth of firm j is:

$$\begin{aligned} W_j(\mathbf{V}; \mathbf{x}_j) &= \sum_{m=1}^2 (p_{jm} - L_{jm} - \text{Exp}_j) q_{jm} - p^{re} Q_j^{re} - \text{FC}_j + \text{ReinsPayout}(\mathbf{V}) \\ &\quad + (1 + r_1) A_{1j} + (1 + r_2) A_{2j} + \text{float}(\mathbf{V}) \\ &= \sum_{m=1}^2 (p_{jm} - L_{jm} - \text{Exp}_j) q_{jm} - p^{re} Q_j^{re} - \text{FC}_j - \\ &\quad \frac{1}{k} \log \left(\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k (\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))} \right) \\ &\quad + (1 + r_1) A_{1j} + (1 + r_2) A_{2j} + \sum_m p_{jm} q(\mathbf{p}_m, \mathbf{R}) \cdot \frac{r_1 A_{1j} + r_2 A_{2j}}{A_{1j} + A_{2j}} \end{aligned}$$

(1) The Price Derivative ($\partial W_j / \partial p_{jm}$):

Applying the product and chain rules, the marginal effect of price on terminal wealth is:

$$\frac{\partial W_j}{\partial p_{j1}} = q_{j1} + \underbrace{\frac{\partial q_{j1}}{\partial p_{j1}}(p_{j1} - L_{j1} - \text{Exp}_j)}_{\gg 0} + \underbrace{\Psi_j}_{\rightarrow 0} L_{j1} \frac{\partial q_{j1}}{\partial p_{j1}} + \underbrace{\Phi_j}_{\text{small}} \left(q_{j1} + p_{j1} \frac{\partial q_{j1}}{\partial p_{j1}} \right) + \underbrace{\frac{\partial W}{\partial R_j} \frac{\partial R_j}{\partial p_{j1}}}_{=0} \quad (18)$$

$$\Psi_j = \frac{\frac{\exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))}{(1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d)))^2}}{\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{p}_m, \mathbf{R}) - d))}}$$

$$\frac{\partial W}{\partial R_j} = \underbrace{\sum_m \frac{\partial q(\mathbf{p}_m, \mathbf{R})}{\partial R_j} (p_{jm} - L_{jm} - \text{Exp}_j)}_{\ll 0} + \underbrace{\Psi_j}_{\rightarrow 0} \sum_m \left(L_{jm} \frac{\partial q(\mathbf{p}_m, \mathbf{R})}{\partial R_j} \right) + \left(\sum_m p_{jm} \frac{\partial q(\mathbf{p}_m, \mathbf{R})}{\partial R_j} \right) \underbrace{\Phi_j}_{\text{small}} \quad (19)$$

The derivative of R_j is found using the Implicit Function Theorem: $\frac{\partial R_j}{\partial x} = \frac{\partial g / \partial x}{1 - \partial g / \partial R_j}$.

$$\frac{\partial g}{\partial p_{j1}} = g_1 \left(\frac{\partial q_1}{\partial p_{j1}} \right) \quad (20)$$

$$\frac{\partial g}{\partial R_j} = g_1 \frac{\partial q_1}{\partial R_j} + g_2 \frac{\partial q_2}{\partial R_j} \quad (21)$$

$$\frac{\partial g}{\partial p_{j2}} = g_2 \left(\frac{\partial q_2}{\partial p_{j2}} \right) \quad (22)$$

In the estimation, $g_1 = 0$. Therefore $\frac{\partial g}{\partial p_{j1}} = 0$ and $\frac{\partial R_j}{\partial p_{j1}} = 0$.

Since $g_2 > 0$ and $\frac{\partial q_2}{\partial R_j} > 0$, then $\frac{\partial g}{\partial R_j} > 0$.

On the boundary, in Equation 18, the second term is massively positive because shed-

ding policies ($\partial q_{j1}/\partial p_{j1} < 0$) eliminates severe catastrophic losses ($p_{j1} - L_{j1} \ll 0$). The fourth term represents the lost investment float on foregone marginal revenue; because $\Phi_j \in [r_1, r_2]$ is a small percentage (e.g., 5%), this term is heavily dwarfed by the second term. Finally, based on derivation and discussion of Equations 20 and 21, the last term is 0. Consequently, $\frac{\partial W_j}{\partial p_{j1}} > 0$.

Similarly, for p_{j2} ,

$$\frac{\partial W_j}{\partial p_{j2}} = q_{j2} + \underbrace{\frac{\partial q_{j2}}{\partial p_{j2}}(p_{j2} - L_{j2} - \text{Exp}_j)}_{\gg 0} + \underbrace{\Psi_j}_{\rightarrow 0} L_{j2} \frac{\partial q_{j2}}{\partial p_{j2}} + \underbrace{\Phi_j}_{\text{small}} \left(q_{j2} + p_{j2} \frac{\partial q_{j2}}{\partial p_{j2}} \right) + \underbrace{\frac{\partial W}{\partial R_j} \frac{\partial R_j}{\partial p_{j2}}}_{> 0} \quad (23)$$

In Equation 23, other terms are similar to discussion of Equation 18. For the last term, $\frac{\partial W}{\partial R_j} < 0$ based on Equation 19. In Equation 22, since $g_2 > 0$, $\frac{\partial q_2}{\partial p_{j2}} < 0$, and $1 - \frac{\partial g}{\partial R_j} > 0$ through calculation, then $\frac{\partial R_j}{\partial p_{j2}} < 0$. Therefore, the last term is positive. This suggests that $\frac{\partial W_j}{\partial p_{j2}} > 0$.

(2) The Reinsurance Derivative ($\partial W_j/\partial Q_j^{re}$):

The marginal payoff of reinsurance, incorporating the rating effect, is:

$$\frac{\partial W_j}{\partial Q_j^{re}} = \underbrace{-p^{re} + \frac{C \exp(-k \cdot C Q_j^{re})}{\exp(-k \cdot C Q_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm} q(\mathbf{P}_m, \mathbf{R}) - d))}}}_{< 0} + \underbrace{\frac{\partial W}{\partial R_j}}_{\ll 0} \underbrace{\frac{\partial R_j}{\partial Q_j^{re}}}_{> 0} \quad (24)$$

As established, even when payout reaches its maximum marginal rate C , empirical estimates show $C < p^{re}$, making the first term in Equation 24 negative when L is large. In the second term, $\frac{\partial W}{\partial R_j} < 0$ and $\frac{\partial R_j}{\partial Q_j^{re}} > 0$. So the second term is also negative. Thus, both terms are negative, ensuring $\frac{\partial W_j}{\partial Q_j^{re}} < 0$.

(3) The Asset Derivatives ($\partial W_j/\partial A_{1j}$ and ∂A_{2j}):

The marginal payoff of holding risk-free capital (A_{1j}) on the boundary is:

$$\frac{\partial W_j}{\partial A_{1j}} = (1 + r_1) + P_{tot} \underbrace{\frac{A_{2j}(r_1 - r_2)}{(A_{1j} + A_{2j})^2}}_{<0} + \underbrace{\frac{\partial W}{\partial R_j}}_{\ll 0} \underbrace{\frac{\partial R_j}{\partial A_{1j}}}_{>0} \quad (25)$$

$$\frac{\partial g}{\partial A_{1j}} = -\frac{\partial g}{\partial \left(\frac{A_{2j}}{A_{1j} + A_{2j}}\right)} \left(\frac{A_{2j}}{(A_{1j} + A_{2j})^2} \right) + \frac{\partial g}{\partial (A_{1j} + A_{2j})} \quad (26)$$

Since in Equation 26, $\frac{\partial g}{\partial A_{1j}} > 0$, then $\frac{\partial R_j}{\partial A_{1j}} > 0$.

In Equation 25, the direct return $(1 + r_1)$ is positive, while the float rotation effect is negative (as $r_1 < r_2$). However, the third term dominates. Holding more capital ex-ante signals safety and increases the firm's credit rating ($\partial R_j / \partial A_{1j} > 0$). In the event of a severe catastrophe, this inflated demand forces the firm to pay out more in claims ($\partial W / \partial R_j \ll 0$). Hence, $\frac{\partial W_j}{\partial A_{1j}} < 0$. By symmetry, in Equation 27, since the third term dominates the second term, the same logic applies to the risky asset, ensuring $\frac{\partial W_j}{\partial A_{2j}} < 0$. In Equation 28, $\frac{\partial g}{\partial A_{2j}} > 0$.

$$\frac{\partial W_j}{\partial A_{2j}} = (1 + r_2) + P_{tot} \underbrace{\frac{A_{1j}(r_2 - r_1)}{(A_{1j} + A_{2j})^2}}_{>0} + \underbrace{\frac{\partial W}{\partial R_j}}_{\ll 0} \underbrace{\frac{\partial R_j}{\partial A_{2j}}}_{>0} \quad (27)$$

$$\frac{\partial g}{\partial A_{2j}} = \frac{\partial g}{\partial \left(\frac{A_{1j}}{A_{1j} + A_{2j}}\right)} \left(\frac{A_{1j}}{(A_{1j} + A_{2j})^2} \right) + \frac{\partial g}{\partial (A_{1j} + A_{2j})} \quad (28)$$

From the three discussions above, I conclude that $\nabla_x W_j \neq 0$.

8.8 NIG distribution

Estimated Parameters:

μ	δ	α	β
0.03366712	0.03188561	7.970371	0.1452077

Notes: This table displays estimated parameters of Normal Inverse Gaussian distribution.

The functional form of the NIG distribution's PDF, denoted as $f(x)$, is given by:

$$f(x|\mu, \alpha, \beta, \delta) = \frac{\alpha\delta}{\pi} e^{\delta\sqrt{\alpha^2-\beta^2}+\beta(x-\mu)} \times \frac{K_1\left(\alpha\sqrt{\delta^2+(x-\mu)^2}\right)}{\sqrt{\delta^2+(x-\mu)^2}}$$

where:

- $\mu \in \mathbb{R}$ is the location parameter.
- $\delta > 0$ is the scale parameter.
- $\alpha > |\beta| \geq 0$ determine the shape of the distribution.
- $K_1(\cdot)$ is the modified Bessel function of the second kind of order 1.

8.9 Hurdle Log-normal Distribution

Estimated Parameters for Hurdle Log-Normal Model			
Variable	p zero (p_0)	meanlog (μ)	sdlog (σ)
loss rate in less risky market	0.6043	-0.5073	0.9992
loss rate in risky market	0.4383	-0.5079	1.0758

Notes: This table displays estimated parameters of Hurdle Log-normal distribution.

The model handles a significant number of zero values (a “hurdle”) and models the positive values using a log-normal distribution. The probability function $f(x)$ is defined as a piecewise function:

$$f(x; p_0, \mu, \sigma) = \begin{cases} p_0 & \text{if } x = 0 \\ (1 - p_0) \cdot \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln(x)-\mu)^2}{2\sigma^2}\right) & \text{if } x > 0 \end{cases}$$

Where:

- p_0 is the probability of observing a zero value.
- The second part is the scaled PDF of the log-normal distribution for positive values.

8.10 Payment Function of Reinsurance

Let the original reinsurance payment function be:

$$f = \min \left\{ CQ_j^{re}, \max \left\{ 0, \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d \right\} \right\}$$

For clarity, define the shorthand variables:

$$A = CQ_j^{re}$$

$$B = \sum_m (L_{jm}q(\mathbf{p}_m, \mathbf{R})) - d$$

Then the function simplifies to:

$$f = \min\{A, \max\{0, B\}\}$$

Using the standard LogSumExp smooth approximations with scaling parameter $k > 0$:

- **Smooth Maximum (Softmax):**

$$\max\{x, y\} \approx \text{softmax}_k(x, y) = \frac{1}{k} \log(e^{kx} + e^{ky})$$

- **Smooth Minimum (Softmin):**

$$\min\{x, y\} \approx \text{softmin}_k(x, y) = -\frac{1}{k} \log(e^{-kx} + e^{-ky})$$

Applying the smooth maximum to $\max\{0, B\}$ gives:

$$\begin{aligned} Y &= \text{softmax}_k(0, B) \\ &= \frac{1}{k} \log(e^{k \cdot 0} + e^{kB}) \\ &= \frac{1}{k} \log(1 + e^{kB}) \end{aligned}$$

Applying the smooth minimum to $\min\{A, Y\}$ gives:

$$\text{softmin}_k(A, Y) = -\frac{1}{k} \log(e^{-kA} + e^{-kY})$$

Substitute $Y = \frac{1}{k} \log(1 + e^{kB})$ into e^{-kY} :

$$\begin{aligned} e^{-kY} &= \exp\left(-k \cdot \left[\frac{1}{k} \log(1 + e^{kB})\right]\right) \\ &= \exp\left(-\log(1 + e^{kB})\right) \\ &= \frac{1}{1 + e^{kB}} \end{aligned}$$

Plugging e^{-kY} into the smooth minimum:

$$\text{softmin}_k(A, Y) = -\frac{1}{k} \log\left(\exp(-kA) + \frac{1}{1 + \exp(kB)}\right)$$

Substituting back $A = CQ_j^{re}$ and $B = \sum_m L_{jm}q(\mathbf{p}_m, \mathbf{R}) - d$, we obtain the final smoothed function:

$$-\frac{1}{k} \log\left(\exp(-k \cdot CQ_j^{re}) + \frac{1}{1 + \exp(k(\sum_m L_{jm}q(\mathbf{p}_m, \mathbf{R}) - d))}\right)$$

References

- A Barbu, D Humphry, and I Sen. The evolution of insurance markets: Insurance provision and capital regulation. Technical report, Working Paper, 2024.
- Christoph Basten and Anastasia Kartasheva. Who monitors climate risk of financial institutions? evidence from catastrophe risks in insurance. *Evidence From Catastrophe Risks in Insurance (May 31, 2024)*. Swiss Finance Institute Research Paper, (24-38), 2024.
- Bo Becker, Marcus M Opp, and Farzad Saidi. Regulatory forbearance in the us insurance industry: The effects of removing capital requirements for an asset class. *The Review of Financial Studies*, 35(12):5438–5482, 2022.
- Steven Berry, James Levinsohn, and Ariel Pakes. Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890, 1995. ISSN 00129682, 14680262. URL <http://www.jstor.org/stable/2171802>.
- Tim J Boonen and Wenjun Jiang. Pareto-optimal reinsurance with default risk and solvency regulation. *Probability in the Engineering and Informational Sciences*, 37(2):518–545, 2023.
- Arthur Charpentier and Benoît Le Maux. Natural catastrophe insurance: How should the government intervene? *Journal of Public Economics*, 115:1–17, 2014.
- Dean Corbae and Pablo D’Erasmus. Capital buffers in a quantitative model of banking industry dynamics. *Econometrica*, 89(6):2975–3023, 2021.
- J David Cummins and Mary A Weiss. Systemic risk and the us insurance sector, 2014.
- J David Cummins, Scott Harrington, and Greg Niehaus. An economic overview of risk-based capital requirements for the property-liability insurance industry. *Journal of Insurance Regulation*, 11(4), 1993.
- Tetiana Davydiuk. Dynamic bank capital requirements. *Available at SSRN 3110800*, 2017.

- Leo De Haan and Jan Kakes. Are non-risk based capital requirements for insurance companies binding? *Journal of Banking & Finance*, 34(7):1618–1627, 2010.
- Evan Eastman and Kyeonghee Kim. Regulatory capital and catastrophe risk. *Available at SSRN 4468543*, 2023.
- Shan Ge. How do financial constraints affect product pricing? evidence from weather and life insurance premiums. *The Journal of Finance*, 77(1):449–503, 2022.
- Shan Ge and Michael S Weisbach. The role of financial conditions in portfolio choices: The case of insurers. *Journal of Financial Economics*, 142(2):803–830, 2021.
- Arnaud Goussebaïle and Alexis Louaas. Catastrophe insurance and solvency regulation. *Available at SSRN 3685086*, 2022.
- Martin F Grace, Scott E Harrington, and Robert W Klein. Risk-based capital and solvency screening in property-liability insurance: hypotheses and empirical tests. *Journal of Risk and Insurance*, pages 213–243, 1998.
- Martin F Grace, Robert W Klein, and Paul R Kleindorfer. Homeowners insurance with bundled catastrophe coverage. *Journal of Risk and Insurance*, 71(3):351–379, 2004.
- Anne Gron. Capacity constraints and cycles in property-casualty insurance markets. *The RAND Journal of Economics*, pages 110–127, 1994.
- Robert W Klein. Insurance regulation in transition. *Journal of Risk and Insurance*, pages 363–404, 1995.
- Robert E. Lucas. Econometric policy evaluation: A critique. *Carnegie-Rochester Conference Series on Public Policy*, 1:19–46, 1976. ISSN 0167-2231. doi: [https://doi.org/10.1016/S0167-2231\(76\)80003-6](https://doi.org/10.1016/S0167-2231(76)80003-6). URL <https://www.sciencedirect.com/science/article/pii/S0167223176800036>.

- Frederic Malherbe. Optimal capital requirements over the business and financial cycles. *American Economic Journal: Macroeconomics*, 12(3):139–174, 2020.
- Greg Niehaus. Managing capital via internal capital market transactions: The case of life insurers. *Journal of Risk and Insurance*, 85(1):69–106, 2018.
- Parinitha Sastry, Ishita Sen, and Ana-Maria Tenekedjieva. When insurers exit: Climate losses, fragile insurers, and mortgage markets. *Fragile Insurers, and Mortgage Markets (December 23, 2023)*, 2023.
- James Stock and Motohiro Yogo. Asymptotic distributions of instrumental variables statistics with many instruments. *Identification and inference for econometric models: Essays in honor of Thomas Rothenberg*, 6:109–120, 2005.
- Johnny Tang. Regulatory competition in the us life insurance industry. 2022.